



Mixed Convection & MHD Effect Analysis in Lid-driven Cavity with Sinusoidally Curved Bottom Wall Using CNT-Water Nanofluid in a Porous Medium

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Declaration of Authorship

We, Md. Mohieminul Islam Khan and Saadbin Khan, declare that this thesis titled, “Mixed Convection & MHD Effect Analysis in Lid-driven Cavity with Sinusoidally Curved Bottom Wall Using CNT-Water Nanofluid in a Porous Medium” and the work presented in it is our own. We confirm that:

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**BANGLADESH UNIVERSITY OF ENGINEERING AND
TECHNOLOGY**

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Abstract

Mixed convection in a lid-driven enclosure with a curved bottom wall has been investigated using CNT (Carbon Nanotube)-water nanofluid in this thesis. The curvature of the bottom wall follows the sine function. Studies have been made with different amplitudes ($\lambda = 0.05, 0.1, 0.15$) of the sine function hence wall curvature. The curved wall at the bottom is heated and the top wall is kept at a relatively low temperature. Left vertical and right vertical surface are assumed to be adiabatic. Top wall has been moving at a constant lid velocity U_0 at right direction. Galerkin method of FEA (Finite Element Analysis) has been used to solve the governing equations. Different parameters like Richardson number ($Ri = 0.1 \sim 10$) at a fixed Reynolds number ($Re = 100$), solid volume fraction of CNT particle ($\phi = 0 \sim 0.09$) are used to observe better heat transfer rate. Streamlines, isothermal lines and average Nusselt number plots are included to discuss the result of the investigation. A 2D plot between average Nusselt number and solid volume fraction of CNT-water nanofluid is also given to analyse heat transfer rate. It is observed that higher value of Richardson number shows better heat transfer rate. Studying these the thesis concludes that better heat transfer is achieved at higher amplitude ($\lambda = 0.15$) of curved surface at higher solid volume fraction ($\phi = 0.09$). Magnetic field is introduced then, which tried to decrease the effect of convection and also decreased flow strength. For that, different Hartmann number ($Ha = 0-100$) is used at a constant $Ri = 70$ and $\phi = 0.09$. And finally combined effect of Darcy number ($Da = 10^{-5} - 10^{-3}$) and Hartmann number is investigated for flow phenomena. Increasing Da causes increment of flow strength.

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Symbols

g	gravitational acceleration	m/s^2
h	heat transfer co-efficient	$\text{W/m}^2\text{K}$
p	fluid pressure	Pa
k	thermal conductivity	W/mK
u	x-axis velocity	m/s
v	y-axis velocity	m/s
x	x-Cartesian coordinate	m
y	y-Cartesian coordinate	m
B_o	magnetic field intensity	Tesla
K	permeability of porous media	m^2
L	length of the cavity	m
P	non-dimensional fluid pressure	
T	fluid temperature	K
T_h	temperature of heated wall	K
T_c	temperature of cold wall	K
U	non-dimensional x-axis velocity	
V	non-dimensional y-axis velocity	
X	non-dimensional horizontal axis	
Y	non-dimensional vertical axis	

greek symbols

α	thermal diffusivity	m^2/s
β	thermal expansion coefficient of nanofluid	$1/\text{K}$
φ	solid volume fraction of CNT-water nanofluid	
ρ	density	kg/m^3
μ	dynamic viscosity	$\text{Pa}\cdot\text{s}$
Θ	non-dimensional fluid temperature	

Subscripts

f	base fluid(water)
s	solid CNT particle
nf	nanofluid

Non-dimensional numbers

Ri	Richardson number
Gr	Grashof number
Ha	Hartmann number
Pr	Prandtl number
Nu	Nusselt number
Da	Darcy number

Dedicated to our beloved parents...

CHAPTER 1

Introduction

When two or more physical systems are in different temperatures, thermal energy is exchanged among them. This phenomenon is known as heat transfer. The rate of heat transfer is dependent on both the temperatures of the systems and the properties of the intervening medium. Heat transfer has a wide range of scope from nano range to bulk range engineering that makes it a pivotal element in an efficient thermal system. It opened an important branch of science that deals with all the causes, consequences and improvement of transfer of thermal energy.

1.1 Fundamental Modes of Heat Transfer

There are three fundamental modes of heat transfer: Conduction, Convection and Radiation.

- Conduction is the transfer of energy through matter from particle to particle. It is the transfer and distribution of heat energy from atom to atom within a substance.
- Convection refers to heat transfer which occurs due to mass movement of substance. This substance could be air, water, any fluid that flows or even some solids like sand.
- Radiation is heat transfer by electromagnetic waves that directly transport energy through space. The special feature of radiation is that it needs no medium.

1.2 Convection Heat Transfer

Heat is transferred from one place to another by bulk fluid motion in convection. Two very important phenomena that account for convection heat transfer are diffusion and advection. Diffusion refers to the random Brownian motion of individual particles in the fluid and advection refers to the process in which heat is transported by the larger-scale motion of currents in the fluid. For example, a heated body diffuses heat from its surface into the fluid and then advection takes place between different layers of fluid. As a result, fluid molecules get heated and move up and relatively cold fluid fills that gap up simultaneously. This molecular process creates a current inside the system that accounts for significant augmentation of heat transfer. This current is known as convection current. Convection heat transfer rate is expressed mathematically by:

$$q = hA(T_a - T_b) \quad (1.1)$$

Here,

q = Heat Transfer Rate

h = Convective Heat Transfer Co-efficient

A = Area of the Object

T_a = Object's Surface Temperature

T_b = Fluid Temperature

Convection heat transfer is classified into three groups: Natural, Forced and Mixed convection heat transfer.

- Natural convection is a convection heat transfer mechanism where the fluid motion is not generated by any external source like a pump, fan, blower or any other suction device.
- Forced convection takes place when a fluid is forced to flow over the surface by an external source such as fans, pumps, suction devices or by stirring, which creates an artificially induced convection current.
- Mixed convection is simply a combination of both natural and forced convection in which both mode of heat transfer occurs simultaneously. Both pressure and buoyant forces react with each other in mixed convection.

1.3 Nanofluids

Nanofluid is fluid containing suspension of nanometer sized particles called nanoparticles. The nanoparticles used in nanofluids are typically made of metals, oxides, carbides, or carbon nanotubes. Their sizes are in the range of 0.1 to 100 nm. Common base fluids include water, ethylene glycol, oil and even air. There are various nanofluids available in the market. Some examples are Cu-water, Al_2O_3 -water, TiO_2 -water, CNT etc.

1.3.1 Heat Transfer Using Nanofluids

Nanofluids are one of most revolutionary inventions in the recent past in the field of thermal engineering. It exhibits excellent thermophysical properties including significant augmentation of thermal conductivity and therefore heat transfer also. It is because of the heat transfer area is increased due to the suspension of nanoparticles. That's why recently it is becoming a very common and popular trend to use nanofluid to enhance heat transfer rate.

1.3.2 CNT-Water Nanofluid

CNT stands for "Carbon Nano Tube". This nanoparticle has been showing great potential in the improvement of thermal performances of fluid in the recent past despite being a little expensive. Basically CNTs are allotropes of carbon with a cylindrical nanostructure. This cylindrical structure contains tubular formation of carbon atoms that have extraordinary mechanical, electrical, thermal, optical and chemical properties. At the individual tube level, these unique structures exhibit 200X the strength and 5X the elasticity of steel, 5X the electrical conductivity, 15X the thermal conductivity and 1000X the current capacity of copper and all these at almost half the density of aluminum. As a carbon based product, CNTs have almost none of the environmental or physical degradation issues common to metals, thermal expansion and contraction, corrosion and sensitivity to radiation, all of which result in greater system failure in performance-sensitive applications. CNTs typically have diameters ranging from less than 1 nm up to 50 nm where a nanometer is one thousand millionth of a meter.

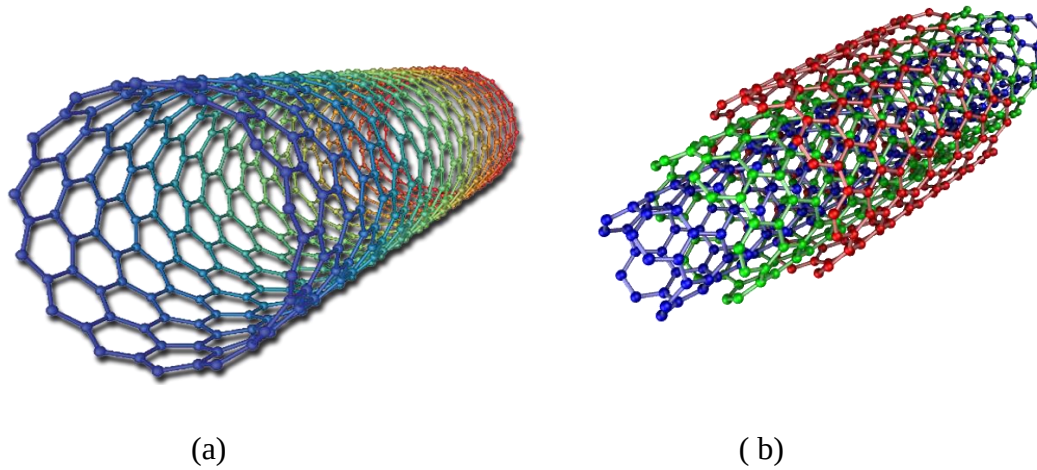


FIGURE 1.1: Structures of (a) Single-walled Nanotube and (b) Multi-walled Nanotube

There are two major types of nanotube structures: Single-walled (SWNT) & multi-walled (MWNT). Single-walled structure has only one rolled layer of graphene whereas multi-walled nanotubes consist of multiple rolled layers. Double-walled carbon nanotubes are one of the common multi-walled forms which exhibits properties that are similar to those of SWNTs but their resistance to chemicals is significantly improved.

1.4 Mixed Convection inside Cavity

Mixed convection in closed cavities has been extensively studied in recent decades. This method of heat transfer has a wide range of applications in engineering. There are a wide and flexible range of shapes for the cavities such as square, triangular, trapezoidal, circular or any other complex shapes. Generally in cavity analysis, walls are kept at different boundary conditions. As example, if there is temperature difference between two walls, there exists a temperature gradient which gives rise to the formation of thermal boundary layer near the walls of the cavity. This also forms a clear density gradient within the fluid layer. Thus the hot fluid leaves the heated surface and the cold fluid fills the void and this process goes on and on resulting in convection current. The characteristics like vortex patterns, isotherms, streamlines etc. of this current vary depending on the shape of the cavity as well as the boundary conditions.

1.5 MHD Convection

MHD stands for Magneto-hydrodynamic. So MHD convection is magneto-hydrodynamic convection, also known as magneto-fluid dynamics. As the name suggests, this is a branch of fluid dynamics where the effect of magnetic field is considered. It is science of the dynamics of matter moving in an electromagnetic field, especially where currents established in the matter by induction modify the field, so that the field and dynamics equations are coupled. It treats, in particular, conducting fluids, whether liquid or gaseous, in which certain simplifying postulates are accepted.

Magneto-hydrodynamic convection has turned out to be a matter of great importance due to its scientific, technological and industrial applications. To keep harmony with its wide range of applications, a considerable amount of research works have been done in this field.

In MHD convection the force applied on fluid is basically a combination of two simultaneous forces, combinedly known as Lorentz force expressed by:

$$\vec{f} = \vec{E} + q(\vec{V} \times \vec{B}) \quad (1.2)$$

Here, $\vec{E}$ represents the force due to the electric field and $q(\vec{V} \times \vec{B})$ is the portion to represent the magnetic force due to the movement of the charged particles where q is the charge of fluid moving with a velocity vector $\vec{V}$ and $\vec{B}$ is the magnetic field applied on the fluid.

1.6 Porosity & Flow through Porous Media

In general, porosity is a measure of the void or empty spaces inside the solid structure or matrix of a material. Porous media is therefore media containing voids or pores which are irregular in shape. Physical form of a porous medium can be compared to a piece of wood or foam. The finer the material of a porous medium is, the greater the specific area than the coarse materials. Based on porosity, the flow quantity and the flow behavior vary significantly. Porosity is proportional to the permeability of a medium. Flow restriction is higher if porosity is less and vice versa.

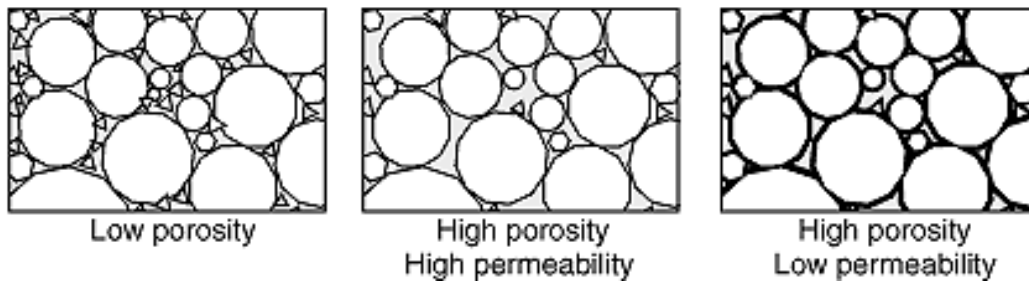


FIGURE 1.2: Effect of Porosity on Permeability

The basic law governing the flow of fluids through porous media is Darcy's law, which was formulated by the French civil engineer Henry Darcy in 1856 on the basis of his experiments on vertical water filtration through sand beds. Darcy found that his data could be described by:

$$Q = \frac{CA\Delta(P - \rho gz)}{L} \quad (1.3)$$

Here,

Q = Volumetric Flow Rate (m^3/s)

C = Constant of Proportionality ($\text{m}^2/\text{Pa.s}$)

A = Cross-sectional Area of Sample (m^2)

P = Pressure (Pa)

ρ = Density (Kg/m^3)

g = Gravitational Acceleration (m/s^2)

z = Vertical Co-ordinate (Measured Downward) (m)

L = Length of Sample (m)

Darcy's law is mathematically analogous to other linear phenomenological transport laws, such as Ohm's law for electrical conduction, Fick's law for solute diffusion, or Fourier's law for heat conduction.

1.7 Motivation of the Present Study

We are living in the era of industrial revolution. Technology has reached to an extent where flexibility of the design of equipments used is almost boundless. But again, in thermal industries the necessity to gain more thermal efficiency is rising in the same pace. So, using nanofluid instead of conventional fluid in various applications as well as applying the advantages of curved or moving heat transfer surfaces can definitely play a vital role in meeting the ever-growing need of high thermal efficiency.

For augmentation of heat transfer performance, nanofluids have become a popular choice for thermal engineers. Numerous nanofluids have been used to investigate the variation of heat transfer and flow performance. It has been found that not all types of nanofluids are suitable for economic application. The nanofluid we used which is CNT-water nanofluid is one of them. But if proper optimization is considered by changing boundary condition and cavity shape, then this nanofluid has got huge potential in thermal application in spite of being a little expensive. These thoughts inspired us to apply curved wall and moving lid technique in the study.

In the last few decades, various experimental and numerical investigations concerning thermal and flow field have been conducted in different types and shapes of cavities. For natural convection, square cavities are the most conventional yet found to have great potential. Our study is also motivated by that suggestion with some significant change of wall shape and boundary conditions only to enhance the required performance.

Magnetic fields generated or applied in practical sectors can cause significant deviation of heat transfer performances. It has been found by previous researchers that magnetic field in system actually decreases the heat transfer rate. So, the effect of Magneto Hydro Dynamics is also shown in the present study. This might help to find a more optimized design to reduce the effect of MHD in future.

Considering the versatility of the mediums used in industries for heat transfer purposes, the effect of using porous medium has also been studied. This extended analysis makes this study more realistic in our point of view.

1.8 Applications of the Present Study

Mixed convection heat transfer is an important phenomenon in engineering systems due to its ever increasing and wide range of applications such as in:

- Lubrication Technologies.
- Electronic Cooling.
- Food Processing.

- Nuclear Reactors.
- Chemical Processing Equipment.
- Solar Energy Collector.

In modern industries designing flexibilities are almost boundless. But the ever increasing requirement of greater thermal efficiency is also a burning question. So inclusion of nanoparticles in complex shaped cavities is under numerous investigations and showing huge potential. Now-a-days many industries are introducing complex heat transfer or cooling equipments with wavy heat transfer surfaces.

Because traditional fluids used for heat transfer applications such as water, mineral oils and ethylene glycol have a rather low thermal conductivity, nanofluids with relatively higher thermal conductivities have attracted enormous interest from researchers due to their potential in the enhancement of heat transfer with little or no penalty in pressure drop.

Effect of MHD has also been very important in the last few years important. We can see its application in:

- Petroleum Industries.
- Plasma Physics.
- Geophysics.
- MHD Pump.
- MHD Flow Meter.
- Cooling of Nuclear Reactors.

Mixed convection with lid-driven-cavities filled with porous media have important application in various fields of engineering and geophysical systems such as:

- Flow and Heat Transfer in Solar Ponds [\[1\]](#)
- Dynamics of Lakes [\[2\]](#)
- Thermal–Hydraulics of Nuclear Reactors [\[3\]](#)
- Electronic Equipment Cooling

1.9 Main Objective of the Investigation

This thesis focuses on the numerical investigation of the flow and thermal field characteristics in a lid driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid considering mixed convection along with the extended effect of MHD and porous medium inclusion. In short, the objectives of the present study are as follows:

- To observe how heat transfer rate is changed for different amplitude of the sinusoidally wavy bottom wall.
- To find the change in average Nusselt number with varying Richardson number.
- To find the effect of different solid volume fractions of nanofluid on average Nusselt number.
- To observe the change in velocity profile for different Richardson number and amplitude of the wavy wall.
- To observe the effect of MHD by varying Hartmann number.
- To observe the effect of porous medium in presence of MHD by varying Darcy number.

1.10 Basic outline of Thesis

Prime target of this thesis is to analyze mixed convection & study the effect of MHD in lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid. An extended investigation of the effect of porous medium has also been conducted.

Chapter 1 of this thesis paper contains the core information on the contents and processes based on which the study is built such as heat transfer modes, three groups of convection processes, brief idea on MHD convection, porous media, nanofluids, its effect etc. This chapter also focuses on the application of our study, inspiration or motivation along with listing the basic objectives of the thesis.

Chapter 2 presents the literature review for the current thesis. These reviews reflect only those previous investigations and their outcome which are based on the physics, geometry and boundary conditions related to the present model we have dealt with.

Chapter 3 is the part where the mathematical model is described based on the physical model. All the dimensional and non-dimensional governing equations are written and relevant boundary conditions are defined properly. Thermal and physical properties of CNT-water nanofluid are also discussed briefly.

Chapter 4 discusses the numerical methods. A brief description of computational fluid dynamics (CFD) is given in the starting of this chapter. Then the finite element method and Galerkin weighted residual method is discussed. To validate our code a previous work has been re-simulated with our code and the outcome is shown here. The grid independency test which was conducted to confirm the numerical accuracy and precision of the study is also discussed in this chapter.

Chapter 5 is all about results and discussion. Effect of all the parameter changes, effect of MHD and porous media inclusion etc. are presented in this chapter with graphical analysis and contours.

Chapter 6 summarizes the present study. All the concluding decisions and further research prospects are briefly discussed here.

CHAPTER 2

Literature Review

The preparation of nanofluid is not an exclusive result of modern research. It has surprisingly long history. Although the technology associated with nanofluid is a fairly new science, but the chief concepts have been developing over the course of fifty years. What has changed recently is our ability to manipulate and engineer in the nanometer scale.

2.1 Nanofluid Heat Transfer

Researches and development of nanofluid in the field of thermal engineering have started not that long ago. The word “nanofluid” was first used by Choi et al. [4] in 1995 who described it as a liquid suspension containing ultra-fine particles with diameter less than 50 nm. Choi gave the first suggestion that nanofluid enhances the thermal conductivity of ordinary fluid. From the experimental work of Eastman et al. [5] it can be seen that an increase in thermal conductivity of approximately 60% can be obtained for a nanofluid consisting of water and 5 vol. % of pure Cu nanoparticles sized less than 10nm. They also showed the particle size effect and potential of nanofluids with smaller particles.

Xuan et al. [6] experimentally investigated flow and heat transfer characteristics for copper-water based nanofluids through a straight tube with a constant heat flux at the wall. Their results show that the nanofluids account for significant augmentation of heat transfer rate compared to pure water. Das et al. [7] also found out that 2 to 4-fold increase in thermal conductivity can be obtained for water-based nanofluids containing Al_2O_3 or CuO. Khanafer et al. [8] studied Cu-water nanofluids in a two dimensional rectangular enclosure. He developed a numerical model to

determine natural convection heat transfer assuming single phase nanofluid. The effect of suspended nanoparticles on buoyancy-driven heat transfer process was analyzed. It was observed that the heat transfer rate increases as the particle volume fraction is increased at any given Grashof number.

2.2 Mixed Convection in Different Shaped Cavities

In last one and a half decade, numerous investigations have been conducted by researchers on change in heat transfer performance and other properties of nanofluid in different geometries of medium and different types and concentration of nanoparticle used. Selvan et al. [9] investigated the mixed convection heat transfer characteristics of a two-sided lid-driven square cavity filled with Cu-water nanofluid. In performing the analysis, it was assumed that the top and bottom moving walls were maintained at different constant temperatures, while the vertical walls were stationary and adiabatic. It was shown that the Cu nanoparticles increased the heat transfer coefficient of the base fluid (water), which eventually results in an improved heat transfer performance.

Arefmanesh et al. [10] conducted a numerical study to examine the effects of uncertainties of viscosity models for the Al_2O_3 –water nanofluid on mixed convection in a square cavity with cold left, right, and top walls and moving hot bottom wall. Their results showed that the average Nusselt number of the hot wall increased with increase in the volume fraction of the nanoparticles for both of used viscosity models. In a numerical study Mahmoodi et al. [11] investigated mixed convection flow of Al_2O_3 –water nanofluid in rectangular cavities with hot moving bottom lid and cold right, left, and top walls. He observed and increase in the rate of the heat transfer via existence of nanoparticles in the base fluid for all range of Richardson number considered.

A benchmark study of thermal conductivity of nanofluids has been published by Bounghirno et al. [12]. This study analyzed the experimental thermal conductivity data gathered by 30 organizations worldwide and found most of them to be consistent within 10%. The study

concludes that the thermal conductivity of a nanofluid increases with the particle concentration and aspect ratio in conformity with the classical Maxwell theory which predicts that the effective thermal ratio k/k_f is a function of particle volume fraction C and the thermal conductivity ratio k_p/k_f . This functional dependence is valid for $C \ll 1$ and $k_p/k_f < 10$. Tzou et al. [13] considered thermal instability of nanofluids in natural convection and concluded that the higher turbulence triggered by the nanoparticles prompted higher heat transfer coefficient than the effect of enhanced thermal conductivity.

Tiwari et al. [14] studied the mixed convection heat transfer performance of Cu-water nanofluid confined within a two-sided lid-driven square cavity with moving vertical walls kept at different constant temperatures. The results showed that the Richardson number and the direction of the moving walls had significant effects on both the fluid flow and the heat transfer within the cavity. It also showed that heat transfer performance is increased when Cu nanoparticles volume fraction is increased in the base fluid. A similar investigation was conducted by Muthamilselvan et al. [15] on rectangular enclosure with different boundary conditions. He also found that increase in the volume fraction of Cu nanoparticles enhanced the heat transfer inside the cavity.

Ben Mansour et al. [16] studied numerically the conjugate problem of laminar mixed convection flow of Al_2O_3 -water nanofluid in a uniformly heated inclined tube. They found that the presence of nanoparticles intensifies the buoyancy-induced secondary flow, especially in the developing region. Their results also show an augmentation of the heat transfer coefficient and a decrease of the wall friction when using nanofluids. For the horizontal inclination, two new correlations were proposed to calculate these two variables in the fully developed region, for Grashof number ranging from 103 to 105 and particle volume concentrations up to 7%.

Talebi et al. [17] studied numerically mixed convection heat transfer in a lid-driven cavity filled with Cu-water nanofluid. The vertical walls of the cavity were differentially heated while its top and bottom walls were kept insulated. Their results showed that an increase in the volume fraction of nanoparticles enhanced the heat transfer inside the cavity. Results of a numerical study on mixed convection flow in an inclined square enclosure filled with Al_2O_3 -water nanofluid were reported by Abu-Nada et al. [18]. They observed significant enhancement in the heat transfer inside the cavity due to the presence of the nanoparticles. Ghasemi et al. [19]

numerically studied the effects of Brownian motion of nanoparticles on the natural convection of a CuO-water nanofluid in a right triangular enclosure with fixed walls. They argued that it was essential to take into account the Brownian motion of nanoparticles in the thermal performance analysis of nanofluid-filled enclosures.

2.3 Some Controversies on the Role of Nanofluids

Most of the above studies have argued that higher thermal conductivity can be achieved in thermal systems utilizing nanofluids. Some researchers, however, argue that the dispersion of nanoparticles in the base fluid may result in a substantial decrease of the heat transfer. Putra et al. [20] conducted an experiment on natural convection of Al_2O_3 -water and CuO-water nanofluids inside a horizontal cylinder heated from one end and cooled from the other. They found a systematic and definite deterioration in natural convective heat transfer. The deterioration increases with particle fraction and appears more significant for CuO-water nanofluid. For a TiO_2 -water nanofluid within a rectangular enclosure heated from below, Wen et al. [21] reported that for the Rayleigh number less than 10^6 , the natural convection heat transfer rate increasingly decreases with the increase of particle fraction, particularly at low Rayleigh numbers.

Incorporating various models for effective thermal conductivity and dynamic viscosity, Hwang et al. [22] theoretically investigated by means of correlation analysis for natural convection heat transfer coefficient of the Al_2O_3 -water nanofluid in a rectangular enclosure heated from below. The effect of the volume fraction on the ratio of heat transfer coefficient of nanofluids to that of base fluid was shown to be dependent on the two models adopted for calculating the effective viscosity of nanofluid. More recently, in a numerical study concerning natural convection in a vertical square enclosure filled with Al_2O_3 -water nanofluid [23], it was shown that significant difference in the effective dynamic viscosity enhancement of the nanofluid calculated from the two adopted formulas, other than that in the thermal conductivity enhancement, could act as a major factor which eventually leads to contradictory results concerning the heat transfer efficacy of using nanofluid in the enclosure.

Nemati et al. [24] studied the mixed convection heat transfer performance of various nanofluids in a square lid-driven cavity with different vertical wall temperatures and a moving upper wall. The results showed that the choice of nanoparticle had a significant effect on the heat transfer performance of the nanofluid. Instead of yielding any direct conclusion, this experiment focused on an interesting new result which was associated with Reynolds number. It was shown that for all of the considered nanofluids, the effect of nanoparticle addition was less significant at higher values of the Reynolds number.

Very recently Sebdani et al. [25] conducted a numerical simulation to investigate effect of variable properties of nanofluid on mixed convection in a square cavity with moving cold side walls and a constant temperature heater on the bottom wall. Their results showed that the heat transfer of the nanofluid could be either enhanced or mitigated with respect to that of the base fluid depending on the Reynolds number and Rayleigh number. They obscurely indicated that the concept of heat transfer and other performance of fluid being always enhanced by adding nanoparticle is not true.

So, it is clear that there is still a controversy on the role of nanofluids in natural convection heat transfer. Modern comprehensive nanofluid simulations investigated by researchers therefore took all the possible properties of the system into considerations which might be viable to affect the ultimate result of the study. These include the structure, shape, size, aggregation and anisotropy of the nanoparticles as well as the type, fabrication process, particle aggregation and deterioration of nanofluids.

2.4 Mixed Convection in Cavity with Wavy Geometries

Beside simple geometries, applications of wavy geometries are increasing day by day such as for enhancing transport performance, industrial use etc. Abu-Nada et al. [26] performed a numerical investigation into the natural convection heat transfer characteristics of Al_2O_3 -water nanofluid in a cavity with wavy walls. Cho et al. [27] investigated the natural convection heat transfer performance of Al_2O_3 -water nanofluid in a complex-wavy wall cavity. Nasrin et al. [28] investigated numerically the influences of physical parameters on mixed convection in a

horizontal lid-driven cavity with wavy base surface. She concluded that the wavy lid-driven cavity can be considered an effective heat transfer mechanism at larger wavy surface amplitude as well as at larger number of waves. Nikfar et al. [29] worked using the same nanofluid with wavy sidewalls. Both study showed that the addition of nanoparticles results in a significant improvement in the mean Nusselt number and thus enhances the rate of convection heat transfer. Moreover, the results showed that the amplitude, wavelength and waveform of the wavy wall have critical effect on the flow and temperature fields thereby they directly affect the heat transfer performance.

2.5 Convection in Cavity with Non-Uniform Temperature Distribution

All the above literatures mentioned above are based on the investigations considering constant temperature or constant heat flux in walls. But in practical cases this exact and accurate thermal boundary conditions are hardly found. So over the past few years some researchers considered the fact that the thermally active walls may be subjected to non-uniform distribution of temperature due to shading or other effects in the fields. As, example we can mention various applied cases such as solar energy collection and cooling of electronic components. Therefore, a large number of studies and numerical investigations have been conducted on both mixed and natural convection heat transfer considering non-uniform temperature distribution on their walls which has been found to be very important in the present scenario of thermal industry.

Sarris et al. [30] in his numerical study, considered an air-filled rectangular enclosure with sinusoidal temperature profile on the upper wall and adiabatic conditions on the bottom and sidewalls. Basak et al. [31] conducted a numerical investigation on laminar natural convection in an air-filled square cavity with both uniformly and non-uniformly heated bottom wall, and adiabatic top wall maintaining constant temperature of cold vertical walls. In his result, he very clearly pointed out the variation in heat transfer performance. Bilgen et al. [32] carried out another numerical study in rectangular enclosures with sinusoidal temperature profiles on side walls and insulated other walls. He considered only natural convection of air in his study. Results of a numerical study on natural convection in an air-filled rectangular enclosure with linear

temperature distributions on both side walls were reported by Sathiyamoorthy et al. [33]. Sivasankaran et al. [34] conducted a numerical study on mixed convection in a lid-driven cavity with sinusoidal temperature distribution on side walls, and moving adiabatic top lid.

2.6 MHD Convection in Cavity

When the fluid is electrically conducting and exposed to a magnetic field the Lorentz force is active and interacts with the buoyancy force in governing the flow and temperature fields. Employment of an external magnetic field has increasing applications in material manufacturing industry as a control mechanism since the Lorentz force suppresses the convection currents by reducing the velocities.

Over the last few years numerous investigations have been conducted on the effect of MHD effect. Oztop et al. [35] studied MHD mixed convection in a lid-driven cavity with corner heater. Oztop et al. [36] also investigated numerical simulation of MHD buoyancy flow in an enclosure when the bottom wall was non-uniformly heated. Rudraiah et al. [37] investigated numerically the effect of magnetic field on natural convection in a rectangular enclosure. They found that the effect of the magnetic field reduced the rate of heat transfer. Maruo et al. [38] numerically investigated the natural convection of a low Prandtl number fluid in the presence of a magnetic field and obtained correlations for the Nusselt number in terms of Rayleigh, Prandtl and Hartmann numbers. Magneto-hydrodynamic natural convection in a vertical cylindrical cavity with sinusoidal upper wall temperature was investigated by Kakarantzas et al. [39]. The authors reported that the increment of Rayleigh number promoted heat transfer by convection. The increment of Hartmann number favored heat conduction and the vertical magnetic field reduced the Nusselt number more than the horizontal.

Rahman et al. [40] investigated MHD natural convection in an enclosure from two semi-circular heaters on the bottom wall. It was reported that the distance between the semi-circular heaters was the most important parameter affecting the heat and fluid flow fields. It was also found that Hartmann number had an adverse effect on heat transfer. Garandet et al. [41] proposed an analytical solution to the governing equations of magneto-hydrodynamics to be used to model

the effect of a transverse magnetic field on natural convection in a two dimensional enclosure. Al-Salem et al. [42] investigated the effects of moving lid direction on MHD convection in a linearly heated cavity with the bottom wall. Hossain et al. [43] analyzed MHD free convection within trapezoidal cavity with non-uniformly heated bottom wall. The authors found out that the average and local Nusselt number at the non-uniform heating bottom wall of the cavity was dependent on the dimensionless parameters and also tilts angles.

Al-Najem et al. [44] used the power law control volume approach to determine the flow and temperature fields under a transverse magnetic field in a tilted square enclosure with isothermal vertical walls and adiabatic horizontal walls at Prandtl number of 0.71 and showed that the suppression effect of the magnetic field on convection currents and heat transfer is more significant for low inclination angles and high Grashof numbers. Sheikholeslami et al. [45] investigated MHD free convection in an eccentric semi-annulus filled with nanofluid. They reported that the heat transfer enhancement increased with increase of Hartmann number and decreased with augment of Rayleigh number.

2.7 Investigations Using CNT

CNT is a relatively expensive nanoparticle comparing to the conventional ones. But it has been showing great potential in thermal engineering in recent times. So, though, there is very few experimental analysis being conducted, numerous numerical investigation are being conducted using this nanoparticle.

Wen and Ding focused on entry region under laminar flow condition using nanofluids containing Al_2O_3 and CNTs in horizontal tube [46, 47]. Ding et al. [48] reported significant enhancement in convective heat transfer of multi-wall carbon nanotube dispersion in water and found that amount of enhancement depends on the Reynolds number, CNT concentration, and pH. Rashidi et al. [49] experimentally investigated heat transfer enhancement of CNT-water nanofluids in a horizontal shell and tube heat exchanger in laminar flow. They found an increase of overall heat transfer coefficient with Reynolds number.

2.8 Mixed Convection in Porous Media

Convection analysis considering porous media has received a great deal of attention. This is due to large number of applications, such as, oil extraction, fluid flow in geothermal reservoirs, solid matrix heat exchangers, iron blast furnaces, energy efficient drying processes, ground water hydrology, solidification of casting etc.

Basak et al. [50] studied numerically the free convection flows in a square cavity filled with a porous matrix for various boundary conditions and with wide range of parameters. Results showed that non-uniform heating of the bottom wall produces greater heat transfer rate at the center of the bottom wall than uniform heating case for all Ra , but average Nu shows overall lower heat transfer rates for non-uniform heating case. Badruddin et al. [51] have studied the steady state heat transfer in a porous medium fixed in a vertical annular cylinder. The Darcy model of flow is employed and heat transfer is assumed to take place by natural convection and radiation. A few recent studies also involve convective heat transfer in triangular cavities. Varol et al. [52] studied free convection in porous media filled right-angle triangular enclosure with insulated vertical wall and differentially heated inclined and bottom walls. They have formulated the flow by Darcy's law and the effect of aspect ratios ranging from 0.25 to 1.0 and Rayleigh numbers from 50 to 1000 has been investigated. They concluded that the maximum mean Nusselt number is obtained for the highest Rayleigh number ($Ra = 1000$) and the lowest aspect ratio ($H/L = 0.25$).

Oztop et al. [53] investigated numerically combined convection heat transfer and fluid flow in a partially heated porous lid-driven cavity. Kandaswamy et al. [54] investigated numerically the mixed convection in a lid-driven square cavity filled with a fluid-saturated porous medium and the results indicate that dominance effects of conduction for low Prandtl numbers and as Prandtl number increases, mixed convection and forced convection dominate the temperature field. The steady mixed convection flow in a vented enclosure with an isothermal vertical wall and filled with a fluid-saturated porous medium is investigated numerically by Mahmud et al. [55]. Volume averaged equations governing unsteady, laminar, mixed convection flow in an enclosure filled with a Darcian fluid-saturated uniform porous medium in the presence of internal heat generation is investigated by Khanafer et al. [56].

2.9 Summary

All these previous studies inspired us to go forward for completing our present thesis. It is clear that mixed convection in a cavity with wavy wall having differential temperature has enormous potential to play a vital role in enhancing thermal efficiency and flow pattern improvement. We also got the idea from these works that the effect of MHD in convection is not negligible and can be used to control the convection rate in proper applications. Lastly we perceived that, consideration of porous medium can fulfill our research and supply a more practical as well as versatile result.

CHAPTER 3

Mathematical Formulation

First step of a numerical investigation is to build a mathematical model, and for doing that a physical model has to be defined properly. Mathematical models are built using mathematical equations. Next step is to solve those equations using relevant boundary conditions which can be found from the physical model. To acquire the most accurate and precise solution of a problem, mathematical modeling is an integral part.

3.1 Physical Model

The geometry considered in this study is a cavity according to figure 3.1

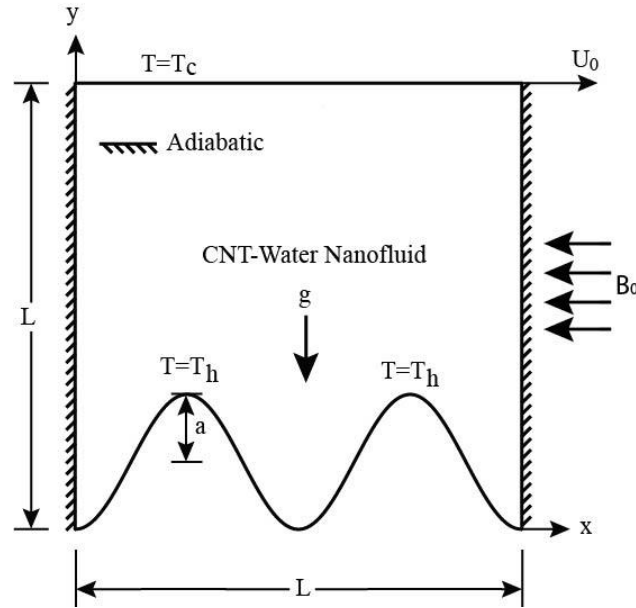


FIGURE 3.1: Schematic Diagram for the Cavity with Wavy Bottom Wall

Width and height of the cavity is denoted by L . The 2-D Cartesian co-ordinate system is defined and the effect of the gravity is shown on the negative y-axis. Top surface of the cavity is moving from left to right horizontally at a constant velocity U_0 . Bottom wall is considered to follow sine function with three different amplitudes 0.05, 0.1 and 0.15 which is denoted by dimensionless parameter $\lambda = a/L$. Top surface is kept at relatively low temperature T_c where the wavy bottom wall is uniformly heated and maintained at T_h . Left and right vertical surfaces are assumed to be adiabatic. A magnetic field B_0 is applied on the system in the horizontal direction. The cavity is filled with CNT-water nanofluid. Nanoparticles and water are assumed to be in thermal equilibrium with no slip between them. The nanofluid is considered Newtonian and incompressible and the flow is assumed to be laminar. The density variation is considered according to the Boussinesq approximation. Other thermo-physical properties of the nanofluid are assumed to be constant. The thermo-physical properties of nanoparticles and water are presented in [Table 3.4](#).

3.2 Mathematical Model

Mathematical model of a system consists of two basic components:

- Governing Equations.
- Proper Boundary Conditions.

Firstly, dimensional governing equations are formulated by studying the physical problems. Then proper boundary conditions are applied to those equations to solve them. After that, the equations and boundary conditions are non-dimensionalized by proper scaling. It should be noted that our studied problem is a single phase problem and we followed Brinkman and Maxwell model to evaluate the thermophysical properties of the nanofluid we used.

3.2.1 Governing Equations

Assumptions have been made that the nanofluid is a Newtonian fluid and the flow is steady laminar flow. Incompressible Navier-Stokes equation has been applied for the two dimensional flow. Flow is assumed to be steady. The density variation is considered according to the Boussinesq approximation. There are three governing equation that we used in our study: Continuity, Momentum and Energy equation. Applying the assumptions stated above the dimensional forms of the governing equations are:

Continuity Equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (3.1)$$

X-momentum Equation:

$$\rho_{nf} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu_{nf} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (3.2)$$

Y-momentum Equation:

$$\rho_{nf} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu_{nf} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \frac{\mu_{nf} u}{K} \quad (3.3)$$

Energy Equation:

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha_{nf} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \quad (3.4)$$

3.2.1.1 Modified Governing Equations for MHD effect

Y-momentum Equation:

$$\rho_{nf} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu_{nf} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + g(\rho\beta)_{nf} (T - T_c) - \sigma B_0^2 v \quad (3.5)$$

X-momentum equation remains the same as we applied the magnetic field horizontally.

3.2.1.2 Modified Governing Equations for Porous Medium and MHD effect

X-momentum Equation:

$$\rho_{nf} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu_{nf} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - \frac{\mu_{nf} u}{K} \quad (3.6)$$

Y-momentum Equation:

$$\rho_{nf} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu_{nf} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + g(\rho\beta)_{nf} (T - T_c) - \sigma B_0^2 v - \frac{\mu_{nf} v}{K} \quad (3.7)$$

3.2.2 Boundary Conditions

To solve the above governing equations proper boundary conditions are applied. For the present problem, bottom wavy wall is heated to a high temperature; top lid is kept at constant low temperature and moving at constant velocity at the right direction. Side walls are adiabatic and stationary. In Table all the dimensional boundary conditions are given.

Boundary wall	Velocity	Temperature
Bottom wavy wall	$u = v = 0$	$T = T_h$
Top horizontal wall	$u = U_0, v = 0$	$T = T_c$
Side wall	$u = v = 0$	$\frac{\partial T}{\partial x} = 0$

TABLE 3.1: Boundary Conditions in Dimensional Form.

3.3 Non-dimensional Analysis

In the present thesis, to solve the governing equations with boundary conditions applied, we followed a popular trend of non-dimensionalizing the equations. This technique exhibits certain advantages over dimensional analysis by reducing physical experiment. Scalings for non-dimensional analysis along with some non-dimensional parameters are described below.

3.3.1 Non-dimensional Scales

Following scales are used to get the non-dimensional governing equation:

$$\begin{aligned}
 X &= \frac{x}{L}, Y = \frac{y}{L}, U = \frac{uL}{\alpha_f}, V = \frac{vL}{\alpha_f} \\
 P &= \frac{(p + \rho_f gy)L^2}{\rho_{nf} \alpha_f^2}, \Theta = \frac{T - T_c}{T_h - T_c} \\
 Ra &= g \beta_f L^3 \frac{T_h - T_c}{\alpha_f \nu_f}, Pr = \frac{\nu_f}{\alpha_f}
 \end{aligned} \tag{3.8}$$

3.3.2 Non-dimensional Governing Parameters

Advantage of non-dimensional analysis is that it reduces the number of experiment required to solve a problem. The non-dimensional parameters don't have any physical dimensions. Following parameters are associated with the present study.

3.3.2.1 Nusselt Number (Nu)

Nusselt number is the ratio of convective to conductive heat transfer across (normal to) the boundary. In this context, convection includes both advection and diffusion. This number is significantly related to heat transfer performances; it was named after Wilhelm Nusselt. It is defined by,

$$Nu = \frac{hL}{k} \quad (3.9)$$

Where, h is the convective heat transfer coefficient of the flow, L is the characteristic length; k is the thermal conductivity of the fluid.

3.3.2.2 Reynolds Number (Re)

The Reynolds number is defined as the ratio of momentum forces to viscous forces and consequently quantifies the relative importance of these two types of forces for given flow conditions. Mathematically it is expressed by,

$$Re = \frac{\rho VL}{\mu} = \frac{VL}{\nu} \quad (3.10)$$

Where,

ρ = Density of the fluid (kg/m^3).

V = Maximum velocity of the object relative to the fluid (m/s).

L = Characteristic linear dimension (m).

μ = Dynamic viscosity of the fluid ($\text{Pa}\cdot\text{s}$)

$\nu = \frac{\mu}{\rho}$ = Kinematic viscosity (m^2/s)

3.3.2.3 Richardson Number (Ri)

Richardson Number expresses the ratio of the buoyancy term to the flow gradient term. It represents the importance of natural convection relative to the forced convection. This term is expressed by,

$$Ri = \frac{g\beta(T_h - T_{ref})L}{V^2} \quad (3.11)$$

Where, g is the gravitational acceleration, β is the thermal expansion coefficient, T_h is the hot wall temperature, T_{ref} is the reference temperature, L is the characteristic length, and V is the characteristic velocity.

The Richardson number can also be expressed by using a combination of the Grashof number and Reynolds number,

$$Ri = \frac{Gr}{Re^2} \quad (3.12)$$

3.3.2.4 Darcy Number (Da)

Darcy number (Da) represents the relative effect of the permeability of the porous medium versus its cross-sectional area. Basically it represents the equivalent or average pore size that the fluid is flowing through. It is defined as,

$$Da = \frac{K}{L^2} \quad (3.13)$$

Where,

- K is the permeability of the medium.
- L is the characteristic length, e.g. the diameter of the particle.

3.3.2.5 Rayleigh number (Ra)

Rayleigh number (Ra) for a fluid is a dimensionless number which indicates ratio of buoyancy and viscosity forces multiplied by the ratio of momentum and thermal diffusivities. Basically it represents the relative magnitude of thermal driving to dissipative forces in the fluid. For natural convection, with the increment Ra , flow changes from laminar to turbulent. It is defined as,

$$Ra = g\beta_f L^3 \frac{T_h - T_c}{\alpha_f \nu_f} \quad (3.14)$$

Where,

ν_f is kinematic viscosity

α_f is the thermal diffusivity.

β_f is the thermal expansion coefficient.

Rayleigh number can also be defined as,

$$Ra = Gr \times Pr \quad (3.15)$$

Here, Gr stands for Grashof number, which is the ratio of the buoyancy to viscous force acting on a fluid.

3.3.2.6 Hartmann Number (Ha)

Hartman number is the ratio of electromagnetic force to the viscous force. In MHD convection, Hartman number becomes a very important parameter. The expression for Hartman number is,

$$Ha = \sqrt{\frac{\sigma_{nf} B_0^2 L^2}{\nu_{nf} \rho_{nf}}} \quad (3.16)$$

where, σ_{nf} is the electrical conductivity of Carbon-Nanotube and B_0 is the magnetic field intensity.

3.3.2.7 Prandtl Number (Pr)

Prandtl Number is a dimensionless number which represents the ratio of momentum diffusivity (kinematic viscosity) to thermal diffusivity and can be expressed as,

$$Pr = \frac{\nu_f}{\alpha_f} \quad (3.17)$$

3.3.3 Non-dimensional Governing Equations

Applying all these dimensionless parameters and proper scaling, a generalized dimensionless form of governing equation is yielded which is:

$$\frac{\partial(U\gamma)}{\partial X} + \frac{\partial(V\gamma)}{\partial Y} = \frac{\partial}{\partial X} \left(\Gamma_\gamma \frac{\partial \gamma}{\partial X} \right) + \frac{\partial}{\partial Y} \left(\Gamma_\gamma \frac{\partial \gamma}{\partial Y} \right) + S_\gamma \quad (3.18)$$

Here non-dimensional dependent variables are designated by γ and corresponding diffusion and source terms are defined by Γ_γ and S_γ respectively. By changing these terms accordingly, all the necessary dimensionless governing equations like continuity, momentum and energy equations can be formed according to [table 3.2](#).

Equation	γ	Γ_γ	S_γ
Continuity	1	0	0
U-momentum	U	$\mu_{nf} / \rho_{nf} \alpha_{nf}$	$-\partial P / \partial X$
V-momentum	V	$\mu_{nf} / \rho_{nf} \alpha_{nf}$	$-\partial P / \partial Y + (\rho\beta)_{nf} RaPr\Theta / (\rho_{nf}\beta_f)$
Energy	Θ	α_{nf} / α_f	0

TABLE 3.2: Summary of the Terms of the Non-dimensional Governing Equations.

3.3.3.1 Modification for MHD Effect

Modified Non-dimensional Y-Momentum Equation for MHD:

$$U \frac{\partial V}{\partial X} + V \frac{\partial V}{\partial Y} = -\frac{\partial P}{\partial Y} + \frac{1}{\text{Re}} \left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2} \right) - \frac{Ha^2}{\text{Re}} V \quad (3.19)$$

X-momentum equation remains the same, because the direction of magnetic field is horizontal.

3.3.3.2 Modification for MHD Effect in Porous Medium

Modified Non-dimensional X-Momentum Equation for MHD in Porous Medium

$$U \frac{\partial U}{\partial X} + V \frac{\partial V}{\partial Y} = -\frac{\partial P}{\partial X} + \frac{1}{\text{Re}} \left(\frac{\partial^2 U}{\partial X^2} + \frac{\partial^2 U}{\partial Y^2} \right) - \frac{1}{\text{Re} Da} U \quad (3.20)$$

Modified Non-dimensional Y-Momentum Equation for MHD in Porous Medium

$$U \frac{\partial V}{\partial X} + V \frac{\partial V}{\partial Y} = -\frac{\partial P}{\partial Y} + \frac{1}{\text{Re}} \left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2} \right) - \frac{Ha^2}{\text{Re}} V - \frac{1}{\text{Re} Da} V + \frac{Gr}{\text{Re}^2} \theta \quad (3.21)$$

3.3.4 Non-dimensional Boundary Conditions

Non-dimensional boundary condition for the present study is presented in [table 3.3](#). This non-dimensional analysis eased the perception of the input and result of the study. Proper scaling was used to achieve these conditions.

Boundary wall	Flow field	Thermal field
Bottom wavy wall	$U = V = 0$	$\Theta = 1$
Top horizontal wall	$U = 1, V = 0$	$\Theta = 0$
Side wall	$U = V = 0$	$\frac{\partial \Theta}{\partial X} = 0$

TABLE 3.3: Boundary Conditions in Non-dimensional Form.

3.4 Thermophysical Properties of the Nanofluid

There are several models to accurately predict the thermophysical properties of nanofluid. Among them, we used Brinkman and Maxwell model to determine the effective dynamic viscosity and effective thermal conductivity respectively. Different other properties are also modeled by the following formulas:

Density:

Density of the nanofluid, ρ_{nf} is dependent on the solid volume fraction, ϕ and the density of the base fluid, ρ_f and can be defined by,

$$\rho_{nf} = (1 - \phi) \rho_f + \phi \rho_s \quad (3.22)$$

Here, the subscripts s, f, nf represents the properties of the nanoparticle, base fluid and nanofluid respectively.

Thermal Diffusivity:

Effective thermal diffusivity of the nanofluid used in the study is quite different from the base fluid diffusivity and is defined by,

$$\alpha_{nf} = k_{nf} / (\rho c_p)_{nf} \quad (3.23)$$

Heat Capacitance:

Heat capacitance of the nanofluid can be found by using the formula,

$$(\rho c_p)_{nf} = (1-\phi) (\rho c_p)_f + \phi (\rho c_p)_s \quad (3.24)$$

Here, c_{pf} and c_{ps} are the heat capacitance respectively of the base fluid and the nanoparticle used.

Thermal Expansion Coefficient:

Thermal expansion coefficient $(\rho\beta)_{nf}$ of the nanofluid can be found by,

$$(\rho\beta)_{nf} = (1-\phi) (\rho\beta)_f + \phi (\rho\beta)_s \quad (3.25)$$

Brinkman Model:

This model was proposed by Brinkman in 1952. It is used to determine the effective dynamic viscosity of the nanofluid and is expressed as follows,

$$\mu_{nf} = \frac{\mu_f}{(1-\phi)^{2.5}} \quad (3.26)$$

This model is limited for the solid volume fraction, ϕ up to 8% only.

Maxwell Model:

Maxwell model has been used to find out the effective thermal conductivity of the nanofluid. This model uses the thermal conductivity of the base fluid and the nanoparticle and is expressed by,

$$\frac{k_{nf}}{k_f} = \frac{k_s + 2k_f - 2\phi(k_f - k_s)}{k_s + 2k_f + \phi(k_f - k_s)} \quad (3.27)$$

In the present experiment, we used CNT-Water Nanofluid to investigate heat transfer and other performances with varying parameters. The main reasons of these types of nanofluids are the improved properties exhibited by the nanofluid. Among numerous types of nanofluids, CNT-Water is showing significant potential in thermal engineering. Thermophysical properties of the nanofluid are presented in [table 3.4](#).

Property	Water	CNT
Specific heat, C_p (J/kg K)	4179	702
Thermal conductivity, k (W/mK)	0.613	3500
Thermal expansion coefficient, β (K^{-1})	21×10^{-5}	0.85×10^{-5}
Density, ρ (kg/m^3)	997.1	1300

TABLE 3.4: Thermophysical Properties of CNT Particles and Water

3.5 Summary

The following topics have been covered in this chapter:

- Physical perception of the study.
- Governing equations have been introduced for mathematical formulation.
- Proper boundary conditions for the study have been defined.
- Both equations and boundary conditions have been non-dimensionalized by dimensionless parameters and proper scaling
- Models for determining effective thermophysical properties of the nanofluid used have been described.

CHAPTER 4

Numerical Simulation

There are experimental, analytical and numerical – these three ways of approaching a physical problem. Numerical approach has become very popular in the last few decades in the field of thermal and fluid engineering. There are several scenarios when numerical analysis becomes the last standing choice for the engineers such as,

- If the problem is very expensive to solve by experimental investigation.
- If millions of iterations are required to solve a problem.
- If physical model of the problem is so complex to perceive completely.

In this chapter, a brief discussion on the numerical method is presented along with the discretization of the method and algorithm followed in the present study.

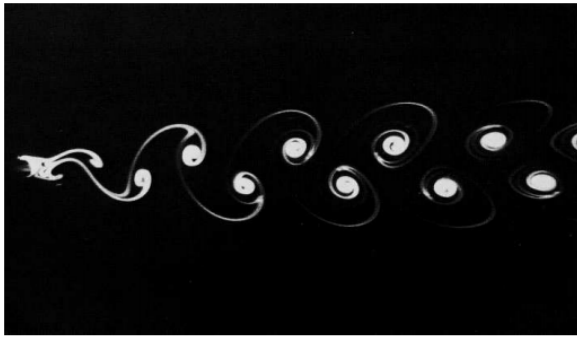
4.1 Basics of Computational Fluid Dynamics (CFD)

CFD is a branch of fluid mechanics that uses numerical methods and algorithm to solve various problems concerning fluid dynamics. This innovative new branch covers a wide range of problems although it is approached in a huge proportion for the solution fluid mechanics related problems. High performance computers are required to perform the calculation.

Basic process of CFD can be classified in three basic categories given below:

- Firstly, preprocessing, where the physical system is modeled mathematically with differential equations, applying suitable boundary conditions to them, selection of solution procedure such as sine difference, finite volume, finite element method etc., discretization of the governing equation, etc.
- Secondly, solving the set of algebraic equations involving boundary conditions by a computer code.
- Finally, post-processing that involves the visualization of dependent variables at every discrete points of the solution domain.

When solution is carried out, the numerical error in each iteration is given the highest priority for the decision making whether to run the next iteration or not. In the investigation of problems related to fluid dynamics and heat transfer, basically Navier-Stokes and energy equation are solved in a CFD problem analysis. The strategy of CFD is to replace the continuous problem domain with a discrete domain using a grid. In the continuous domain, each flow variable is defined at every point in the domain. In the discrete domain, each flow variable is defined only at some the definite grid points.



real experiment



CFD simulation

FIGURE 4.1: Difference between Real Experiment and CFD Simulation

Computational fluid dynamics constitutes a relatively new approach in the philosophical study and development of the whole discipline of fluid dynamics. The advent of the high speed digital computer combined with the development of accurate numerical algorithms for solving physical problems on these computers has revolutionized the way we practice fluid dynamics today.

Computational fluid dynamics is today an equal partner with pure theory and pure experiment in the analysis and solution of fluid dynamics problems. There is no doubt that computational fluid dynamics will continue to play this role indefinitely.

4.2 Applications of CFD

CFD is useful in a wide variety of applications and here we note a few to give you an idea of its use in industry.

- CFD can be used to simulate the flow over a vehicle. For instance, it can be used to study the interaction of propellers or rotors with the aircraft fuselage.
- The CFD analysis showed the effectiveness of a simpler manifold design without the need for field testing.
- Bio-medical engineering is a rapidly growing field and uses CFD to study the circulatory and respiratory systems.
- CFD is attractive to industry since it is more cost-effective than physical testing. However, one must note that complex flow simulations are challenging and error-prone and it takes a lot of engineering expertise to obtain validated solutions.

4.3 Finite Element Method (FEM)

The finite volume method is a method for representing and evaluating partial differential equations in the form of algebraic equations. Similar to the finite difference method or finite element method, values are calculated at discrete places on a meshed geometry. "Finite volume" refers to the small volume surrounding each node point on a mesh. In the finite volume method, volume integrals in a partial differential equation that contain a divergence term are converted to surface integrals, using the divergence theorem. These terms are then evaluated as fluxes at the surfaces of each finite volume. Because the flux entering a given volume is identical to that

leaving the adjacent volume, these methods are conservative. Another advantage of the finite volume method is that it is easily formulated to allow for unstructured meshes. The method is used in many computational fluid dynamics packages.

At the initial stage of FEM the problem geometry is discretized into several elements. The sizes and shapes of the element can vary. As example, it is very common to use quadratic element, triangular element, etc. These mesh elements have certain number of nodes. The governing equations are discretized at every two consecutive nodes at different approaches. . Some of the most common approaches are least square methods, point collection method, Galerkin weighted residual method, etc. An approximate solution and weighting function are assumed for obtaining the solution matrix. Approximation of solution determines the success of the method. The higher the order of the polynomial approximated, the closer the solution is to the exact solution. Ultimate target of Finite Element Method is to reduce the residual error to a preset tolerable limit.

4.3.1 Galerkin Weighted Residual Method of FEA

In our study we used the Galerkin Weighted Residual Method of FEA. In this method, the approximated solution and the weighted function are the same. The governing equations are solved applying specified boundary conditions using parametric solver. All the dependent variables like pressure, velocity and temperature are calculated at each node with the specific boundary conditions. They form a set of linear or non-linear algebraic equations which solved using iterative technique. We used a residual tolerance criterion of 10^{-6} in our approach. The solution converges when the difference of the dependent variable in successive iteration is within this range of convergence.

4.4 Discretization of Equations

For the present problem six noded triangular mesh element is chosen. The elements are of arbitrary triangular shape and non-uniform in nature. These elements assume linear interpolation for the velocity components, the pressure, and the temperature as,

$$U(X, Y) = N_i U_i \quad (4.1)$$

$$V(X, Y) = N_i V_i \quad (4.2)$$

$$P(X, Y) = N_i P_i \quad (4.3)$$

$$\Theta(X, Y) = N_i \Theta_i \quad (4.4)$$

Where, $i = 1, 2, 3, 4, 5$ & 6 which corresponds the node number for a particular elements and the element interpolation functions are represented by N_i . Final objectives of analysis are to solve momentum equations for both dimensions to obtain velocity components and using continuity equation to solve pressure field in entire domain and use energy equation for solving the temperature.

4.4.1 Discretization of the Momentum Equations

The two momentum equations found from equation (3.18) and table 3.2 are discretized using the conventional Galerkin method. However, a special treatment of the convection terms is incorporated. Using the standard Galerkin approach, each momentum equation is multiplied by weighting functions, N_i and then the diffusion terms are integrated by parts over the element domain using Gauss theorem to yield the finite element equations in the form,

$$AU = R_{pX} + R_U + R_a \quad (4.5)$$

$$AV = R_{pY} + R_V + R_b \quad (4.6)$$

Where, the coefficient matrix A contains the known contributions from the convection and diffusion terms. The load vectors on the right-hand side of (4.5) and (4.6) are defined by,

$$R_{PX} = - \int_{\Omega} N_i \frac{\partial P}{\partial X} d\Omega \quad (4.7)$$

Where, R_{PX} is the residual for the pressure term working in the x-direction.

$$R_{PY} = - \int_{\Omega} N_i \frac{\partial P}{\partial Y} d\Omega \quad (4.8)$$

Where, R_{PY} is the residual for the pressure term working in the y-direction.

$$R_U = \frac{\nu_{nf} \text{Pr}}{\nu_f} \int_{\Gamma} N_i \left(\frac{\partial U}{\partial X} n_x + \frac{\partial U}{\partial Y} n_y \right) d\Gamma \quad (4.9)$$

Where, R_U is the residual term for the velocity component acting on the x-direction.

$$R_V = \frac{\nu_{nf} \text{Pr}}{\nu_f} \int_{\Gamma} N_i \left(\frac{\partial V}{\partial X} n_x + \frac{\partial V}{\partial Y} n_y \right) d\Gamma \quad (4.10)$$

Where, R_V is the residual term for the velocity component acting on the y-direction.

$$R_a = \int_{\Omega} N_i \frac{\text{Pr}}{Da} \frac{\nu_{nf}}{\rho_{nf} \nu_f} U d\Omega \quad (4.11)$$

Where, R_a is the residual term for the body force component acting on the x-direction for the buoyancy force, magnetic field and porous media.

$$R_b = \int_{\Omega} N_i \frac{\text{Pr}}{\text{Da}} \frac{\nu_{nf}}{\rho_{nf} \nu_f} V d\Omega \quad (4.12)$$

Where, R_b is the residual term for the body force component acting on the y-direction for the buoyancy force, magnetic field and porous media. Ω is the element area and r is the element boundary.

4.4.2 Discretization of the Pressure Equation

To derive discretized pressure equation, the method of weighted residuals is applied to the continuity equation found from equation 3.18,

$$\int_{\Omega} N_i \left(\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} \right) d\Omega = - \int_{\Omega} N_i \left(\frac{\partial N_i}{\partial X} U + \frac{\partial N_i}{\partial Y} V \right) d\Omega + \int_{\Gamma} N_i (U n_x + V n_y) d\Gamma \quad (4.13)$$

Where, the integrations are performed over the element domain Ω and along the element boundary Γ ; n_x and n_y are the direction cosines of the unit normal to the element boundary with respect to x and y-directions, respectively. Now it can be considered that,

$$A_{ii} U_i = - \sum_{j \neq i} A_{ij} U_j + f_i^U - \int_{\Omega} N_i \frac{\partial P}{\partial X} d\Omega \quad (4.14)$$

$$A_{ii} V_i = - \sum_{j \neq i} A_{ij} V_j + f_i^V - \int_{\Omega} N_i \frac{\partial P}{\partial Y} d\Omega \quad (4.15)$$

Where, f_i^U and f_i^V are the surface integral and the source term due to buoyancy in the x and y-directions, respectively. By assuming constant pressure gradient on an element it can be found that,

$$U_i = \bar{U}_i - K_i^P \frac{\partial P}{\partial X} \quad (4.16)$$

$$V_i = \bar{V}_i - K_i^P \frac{\partial P}{\partial Y} \quad (4.17)$$

Where,

$$\bar{U}_i = \frac{1}{A_{ii}} \left(-\sum_{j \neq i} A_{ij} U_j + f_i^U \right) \quad (4.18)$$

$$\bar{U}_j = \frac{1}{A_{ii}} \left(-\sum_{j \neq i} A_{ij} V_j + f_i^V \right) \quad (4.19)$$

$$K_i^P = \frac{1}{A_{ii}} \left(\int_{\Omega} N_i d\Omega \right) \quad (4.20)$$

By substituting the element velocity interpolation functions, (4.1) and (4.2) in the continuity equation, (4.13) takes the form as,

$$\int_{\Omega} \frac{\partial N_i}{\partial X} N_j U_j d\Omega - \int_{\Omega} \frac{\partial N_i}{\partial Y} N_j V_j d\Omega + \int_{\Gamma} N_i (U n_x + V n_y) d\Gamma = 0 \quad (4.21)$$

Introducing the nodal velocities U_i and V_i from (4.19) and (4.20) in (4.21) yields,

$$\begin{aligned} \int_{\Omega} \frac{\partial N_i}{\partial X} (N_j K_j^P) \frac{\partial P}{\partial X} d\Omega + \int_{\Omega} \frac{\partial N_i}{\partial Y} (N_j K_j^P) \frac{\partial P}{\partial Y} d\Omega &= \int_{\Omega} \frac{\partial N_i}{\partial X} (N_j \bar{U}_j) d\Omega \\ &+ \int_{\Omega} \frac{\partial N_i}{\partial Y} (N_j \bar{V}_j) d\Omega - \int_{\Gamma} N_i (U n_x + V n_y) d\Gamma \end{aligned} \quad (4.22)$$

Finally, element pressure interpolation functions are applied to (4.22) and can be written in matrix form with unknown of the nodal pressure as,

$$(K_X + K_Y)P = F_U + F_V + F_C \quad (4.23)$$

Where,

$$K_X = \int_{\Omega} \frac{\partial N_i}{\partial X} (N_j K_j^P) \frac{\partial N_k}{\partial X} d\Omega \quad (4.24)$$

$$K_Y = \int_{\Omega} \frac{\partial N_i}{\partial X} (N_j K_j^P) \frac{\partial N_k}{\partial Y} d\Omega \quad (4.25)$$

$$F_U = \int_{\Omega} N_j \bar{U}_j \frac{\partial N_k}{\partial X} d\Omega \quad (4.26)$$

$$F_V = \int_{\Omega} N_j \bar{V}_j \frac{\partial N_k}{\partial Y} d\Omega \quad (4.27)$$

$$F_e = - \int_{\Gamma} N_i (U n_x + V n_y) d\Gamma \quad (4.28)$$

Above element pressure equations are tied to form the global equations boundary conditions for the specified nodal pressures are imposed prior to solving for the updated nodal pressures.

4.4.3 Discretization of the Energy Equation

The discretization of the energy equation is performed in the similar way of the momentum equations. The standard Galerkin weighted residuals method is applied to yield the element equations, which can be written in stiffness matrix form as,

$$K\Theta = R \quad (4.29)$$

Where, the matrix K consists of the known contributions from the convection and diffusion terms and the load vector R represent the heat flux along the element boundary as follows:

$$R = \frac{\alpha_{nf}}{\alpha_f} \int_{\Gamma} N_i \left(\frac{\partial \Theta}{\partial X} n_x + \frac{\partial \Theta}{\partial Y} n_y \right) d\Gamma \quad (4.30)$$

These element equations are again assembled to yield the global temperature equations. Appropriate boundary conditions are applied prior to solving for the new temperature values at every node.

4.5 Calculation of Streamfunction

Assuming the flow is incompressible and irrotational, the relation of streamfunction (Ψ) with velocity component can be expressed as,

$$U = \frac{\partial \Psi}{\partial Y}, V = -\frac{\partial \Psi}{\partial X} \quad (4.31)$$

For obtaining the streamfunction value, it is needed to solve the Poisson's equation. The Poisson's equation can be expressed as,

$$\frac{\partial^2 \Psi}{\partial X^2} + \frac{\partial^2 \Psi}{\partial Y^2} = \frac{\partial U}{\partial Y} - \frac{\partial V}{\partial X} = -\zeta \quad (4.32)$$

Where, right hand term of the equation comes from the velocity (ζ).

For solving the Poisson's equation, the boundary conditions are for all walls: $\Psi = 0$.

4.6 Calculation of the Performance of Heat Transfer

Important non-dimensional parameters such as average fluid temperature inside the cavity can be calculated using the following expression.

$$\Theta_{av} = \frac{1}{A} \int_A \Theta dA \quad (4.33)$$

Where, A is the non-dimensional area of the cavity given by,

$$A = \left(a - \frac{ab}{L^2}\right) \quad (4.34)$$

The characteristics of heat transfer can be evaluated using average Nusselt number of the heated wall and is expressed by the following relation,

$$Nu = \frac{hL}{k_f} = -\frac{k_{nf}}{k_f} \int_0^1 \left(\frac{\partial \Theta}{\partial Y} \right)_{X=0} dY. \quad (4.35)$$

4.7 Mesh Generation

For the present study, six-noded triangular mesh arrangement is applied for the discretization of the present problem domain. The entire domain of the geometry is discretized using non-uniform triangular mesh arrangement like figure 4.2. The domain near the boundaries is meshed using more fine triangular elements as the boundary layers form in the region. Also the meshing is done in such a manner as that there is no singular point in the computational domain.

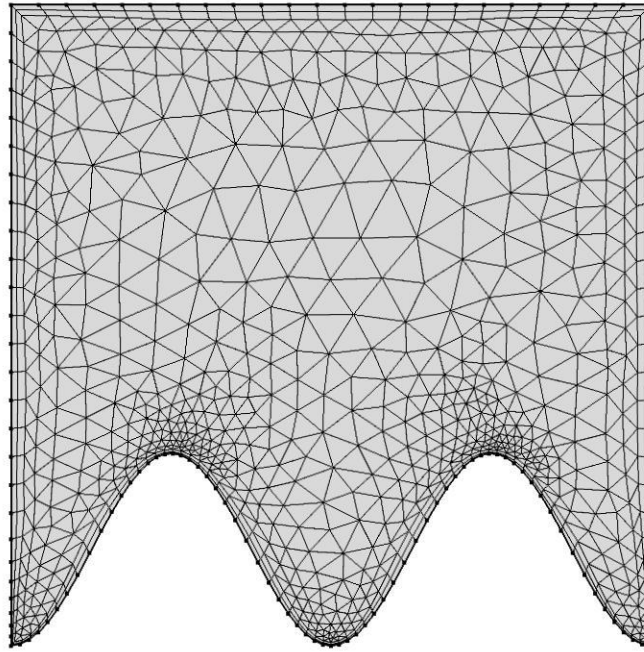


FIGURE 4.2: Triangular Mesh Generated in Cavity with Wavy Wall

4.8 Grid Independency Test

The system considered in a simulation is divided into many cells, which are called grids. And it is assumed that all the elements in one grid area have the same conditions and characteristics. But in real, a system is made of infinite number of elements. So, the finer and larger the number of grids that is used in a simulation, the closer the simulation will be to the real solution. Grid convergence is the term used to describe the improvement of results by using successively smaller cell sizes for the calculations. A calculation should approach the correct answer as the mesh becomes finer, hence the term grid convergence. Though using smaller size of grids increases grid number hence gives more accurate answer, it increases the time of calculation also. And this is not desirable. So in order to overcome this dilemma a CFD technique is used to start with a coarse mesh and gradually refine it until the changes observed in the results are smaller than a pre-defined acceptable error. This approach is called grid-independency test.

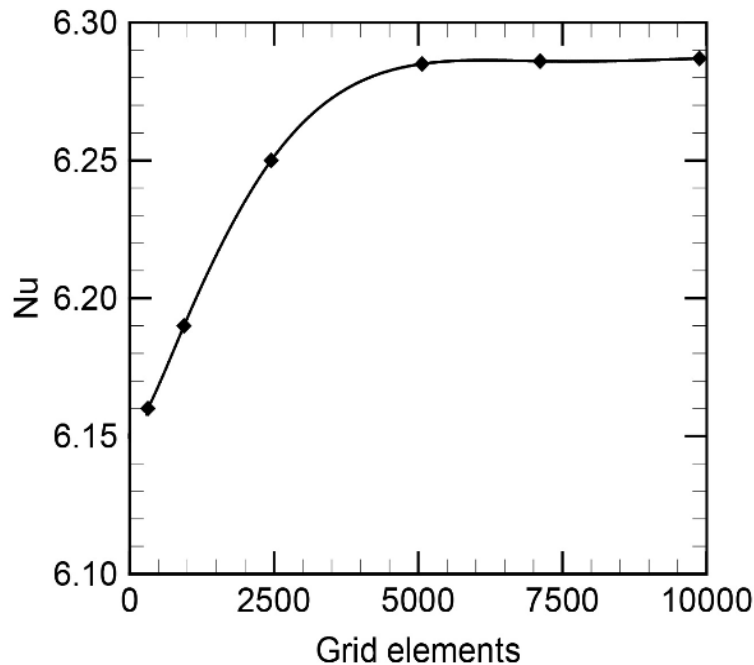


FIGURE 4.3: Variation of Nusselt Number with Different Grid Elements at $Ri = 1$, $Re = 100$, $\phi = 0.1$ for $\lambda = 0.15$.

Since numerical results are greatly depends on the mesh generation, a grid-independence test is performed to find out the optimum number of elements to discretize the computational domain. A further increment of grid number will give better results to a small percent but at a cost of high computational time and memory, which makes an certain error acceptable. Extensive tests are carried out with different sizes of mesh number. For grid independency test, variation of average Nusselt number is investigated considering mixed convection for six different number (314, 940, 2452, 5067, 7114 and 9875) of grid elements at $Ri = 1$, $Re = 100$, $\phi = 0.1$ and $\lambda = 0.15$. The results are shown in Fig.1. From number of elements 5067 no mentionable variation of Nu is observed. Further improvement of mesh size does not result in any significant change in Nusselt number. Therefore, a number of 5067 elements is selected as the ideal mesh size.

4.9 Summary

The focus of this chapter was to describe finite element formulation using Galerkin weighted residual method for the present work. Also discretization of the governing equations is presented and numerical methods of solving are discussed elaborately. A grid independency test is carried out to discuss the numerical accuracy.

CHAPTER 5

Results and Discussion

The study was done to find out the effect of change in curvature amplitude of the heated bottom wall, Richardson number, Hartmann number and Darcy number on mixed convection inside a lid driven cavity. The cavity was filled with CNT-water nanofluid. By varying Hartmann number and Darcy number basically the effect of different magnetic field and effect of porosity change was studied respectively. Also investigating the effect of solid volume fraction (up to 9%) on heat transfer is a part of the investigation. Effect of various curvature amplitude of the bottom wall was studied by varying the corresponding parameter λ . Three different values taken for λ are 0.05, 0.10, and 0.15. Influence of Richardson number is observed by varying its value from 0 to 10. Effect of solid volume fraction ϕ on heat transfer was investigated by observing average Nusselt number. Then for a certain Richardson number and ϕ effect of different magnetic field on isotherms and streamlines was studied by varying Hartmann number from 0 to 100. Beside isotherms, effect of magnetic field on heat transfer was further investigated by studying average Nusselt number too. And finally combined effect of various Darcy number and Hartmann number on streamlines was studied in order to find out the contribution of porous media and magnetic field.

5.1 Effect of Amplitude on Streamline and Isotherm

From [figure 5.1](#), by observing the streamlines for $Ri = 1$ and various sinusoidal wavy surface wall amplitude (λ) it can be seen that single large fully developed clockwise eddy is formed inside the cavity due to nanofluid.

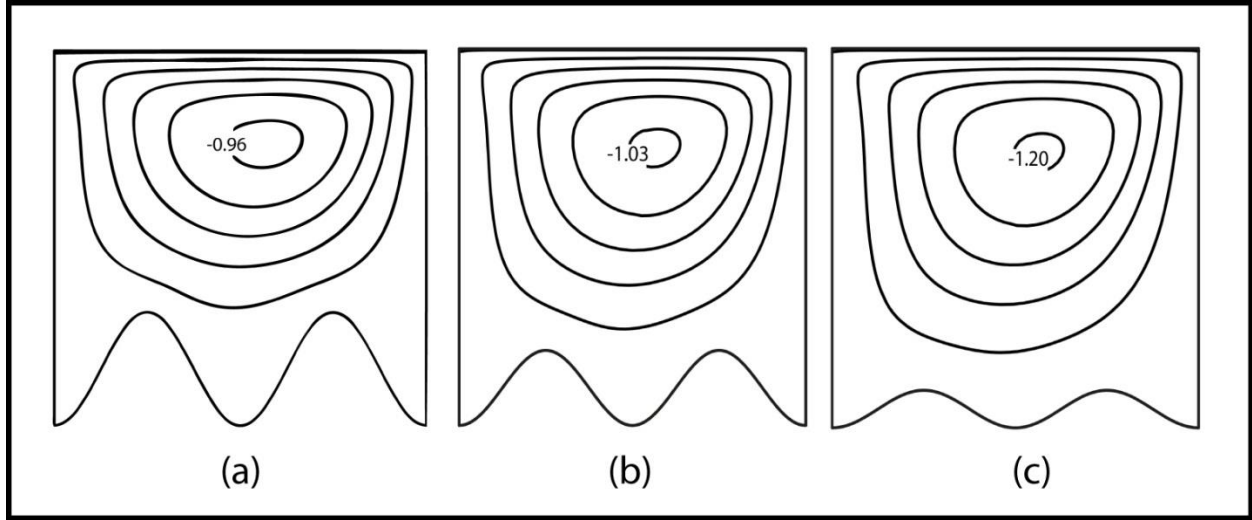


FIGURE 5.1: Comparison of Streamline Contours for (a) $\lambda = 0.15$; (b) $\lambda = 0.10$; (c) $\lambda = 0.05$ for $Ri = 1$, $\varphi = 0.09$ and $Gr = 1000$.

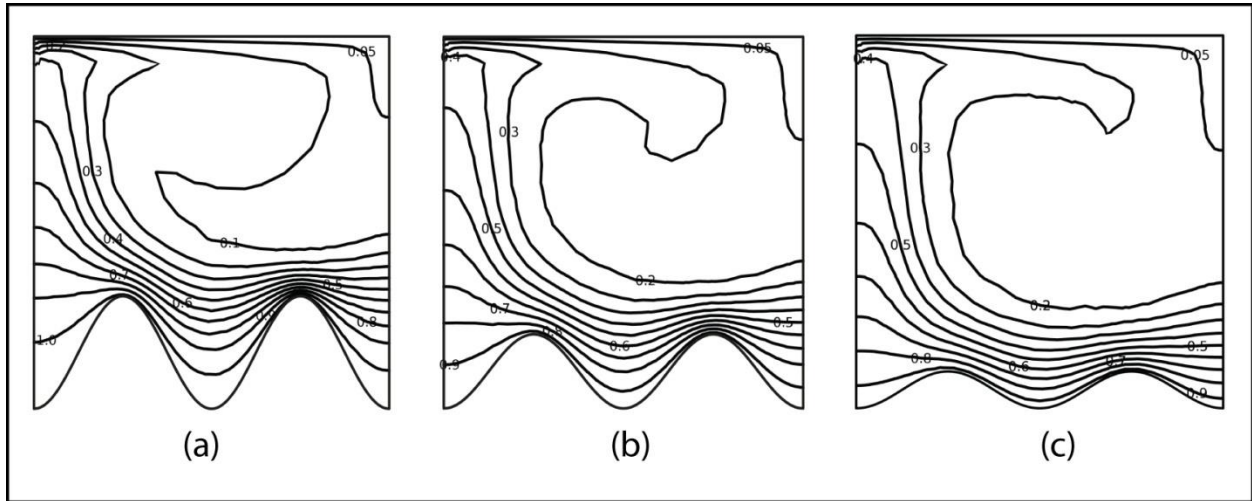


FIGURE 5.2: Comparison of Isotherm Contours for (a) $\lambda = 0.15$; (b) $\lambda = 0.1$; (c) $\lambda = 0.05$ for $Ri = 1$, $\varphi = 0.09$ and $Gr = 1000$.

Based on the amplitude of the wavy wall the core position varies. As $Gr = 1000$, buoyancy force cannot overcome the effect of viscous force. Hence the flow inside the cavity is basically established by the moving top surface. That is the reason why the location of the core of the eddy is more near to the moving top wall than the bottom wall. From [figure 5.2](#), by observing the isotherms for a constant Ri and various λ , it can be observed that the isotherms are bundled at a greater extent near the bottom surface where the gathering rather quiet light in the interior of the cavity. Thus it indicates that a steep temperature gradients in the vertical direction in this part of the cavity, while the inner part has a very weak temperature gradients. When $\lambda = 0.05$ the isotherms cluster more into the peak region of the curve of the bottom surface. It is due to the cavity area reduction. For the two other cases ($\lambda = 0.1$ and 0.15) the heavy cluster areas are akin at the peak areas.

5.2 Analysis of Mid-plane Vertical Velocity

[Figure 5.3](#) represents mid-plane vertical velocity, V along the x -location of the cavity. It was taken for different Richardson Number and different amplitude of wavy wall (λ). The figure indicates that the velocity increases close to the cold surface at left. It is expected due to thermal buoyancy effect as the wall creates the most temperature difference. Figure shows that when Ri is lowest, amplitude of velocity is higher at high λ . At $Ri = 0.1$ velocity is higher for $\lambda = 0.15$ than $\lambda = 0.05$, though the difference is not much mentionable. But as Richardson number goes up, buoyancy force becomes more dominant and amplitude of vertical velocity gradient is decreased. At $Ri = 10$ velocity amplitude is notably high for $\lambda = 0.05$ than $\lambda = 0.15$.

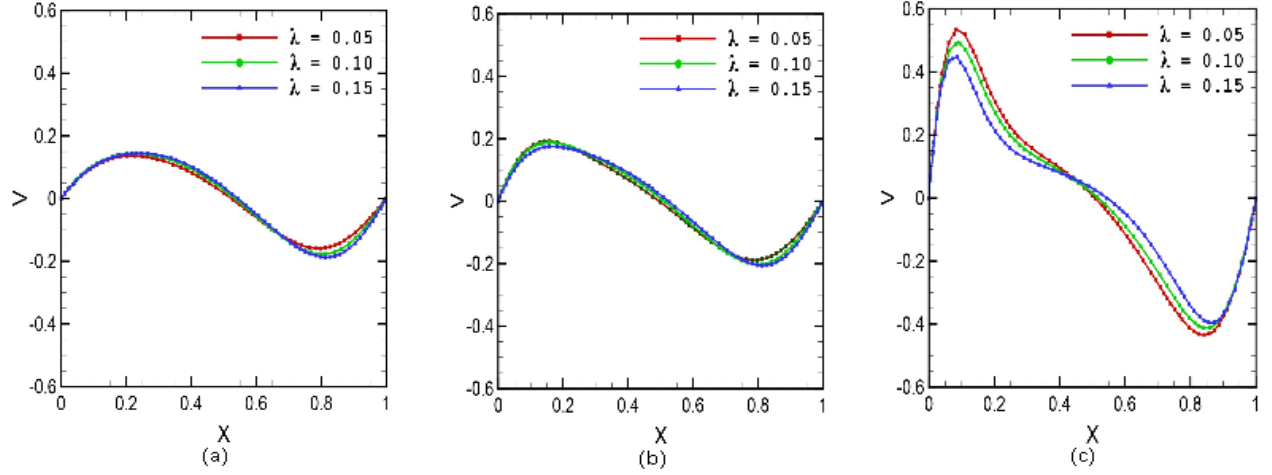


FIGURE 5.3: Comparison of Horizontal Mid-plane Velocity at Different λ for (a) $Ri = 0.1$; (b) $Ri = 1$; (c) $Ri = 10$.

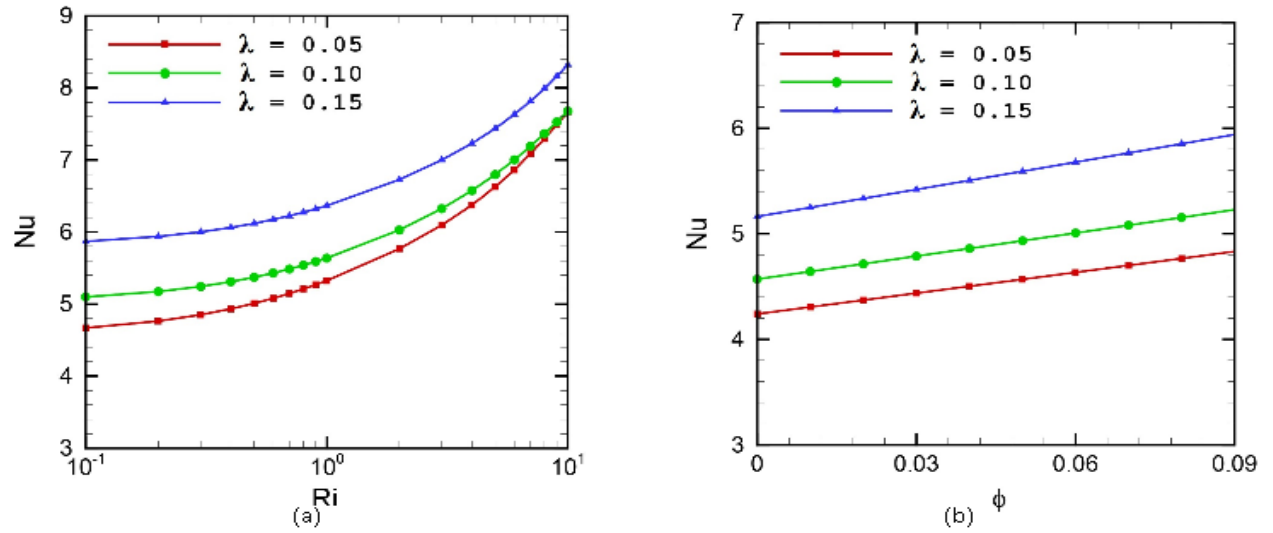


FIGURE 5.4: Variation of Average Nusselt Number for Different λ with (a) Richardson Number Ri and (b) Solid Volume Fraction ϕ .

5.3 Effect of Different Parameters on Average Nusselt Numbers

Figure 5.4 presents the change of average Nusselt number, Nu for volume fraction $\varphi = 0.09$ with Richardson number, Ri for different λ . From the figure it can be clearly seen that Nu increases exponentially with the increment of Ri . At a same Ri , average Nusselt number is different for different λ . Nu is higher for higher λ ($= 0.15$) whereas Nu is lowest for the minimum λ ($= 0.05$). It is also seen that with the increment of Ri , increment rate of Nu increases as the value of λ goes down. As a result the amplitude of Nu difference decreases with Ri increase. Fig. 7(b) shows the variation of Nu for a constant Ri ($=1$) at φ . It is evident that, Nu increases linearly with the increment of volume fraction at any λ . As the volume fraction, φ is the measurement of nanoparticles quantity in nanofluid it can be said that with the increment of nanoparticle volume Nusselt number increases. Apparently at a certain φ , average Nusselt number is maximum for stronger wave amplitude ($\lambda = 0.15$) and lower as the amplitude gets lower. It is minimum at $\lambda = 0.05$. Also the difference between the amplitudes of Nu decreases with λ value reduction.

5.4 Effect of Different Hartmann Numbers on Streamlines

Hartmann number corresponds to the magnetic effect applied on the cavity. Increasing Hartmann number means strengthening applied magnetic field. For a certain value of $\varphi = 0.09$ and $Ra = 10^6$ effect of Hartmann number was studied. Figure 5.5 shows the effect of increasing Hartmann number on the streamlines of the cavity. When $Ha = 0$, no magnetic field is applied at all, streamlines give a natural lid-driven phenomena of velocity field. But with the increment of Ha strength of flow starts decreasing. Introduction of magnetic field tends to retard the motion of the cavity. So, flow at the middle section starts to become motionless, effect of driven lid on streamlines starts to lose its significance. For the maximum $Ha = 90$, Figure 5.5 (d) suggests that flow loses its strength so much that the circulation due to driven lid can't even reach to the bottom of the cavity. Eddy core moves upward to the driven top lid.

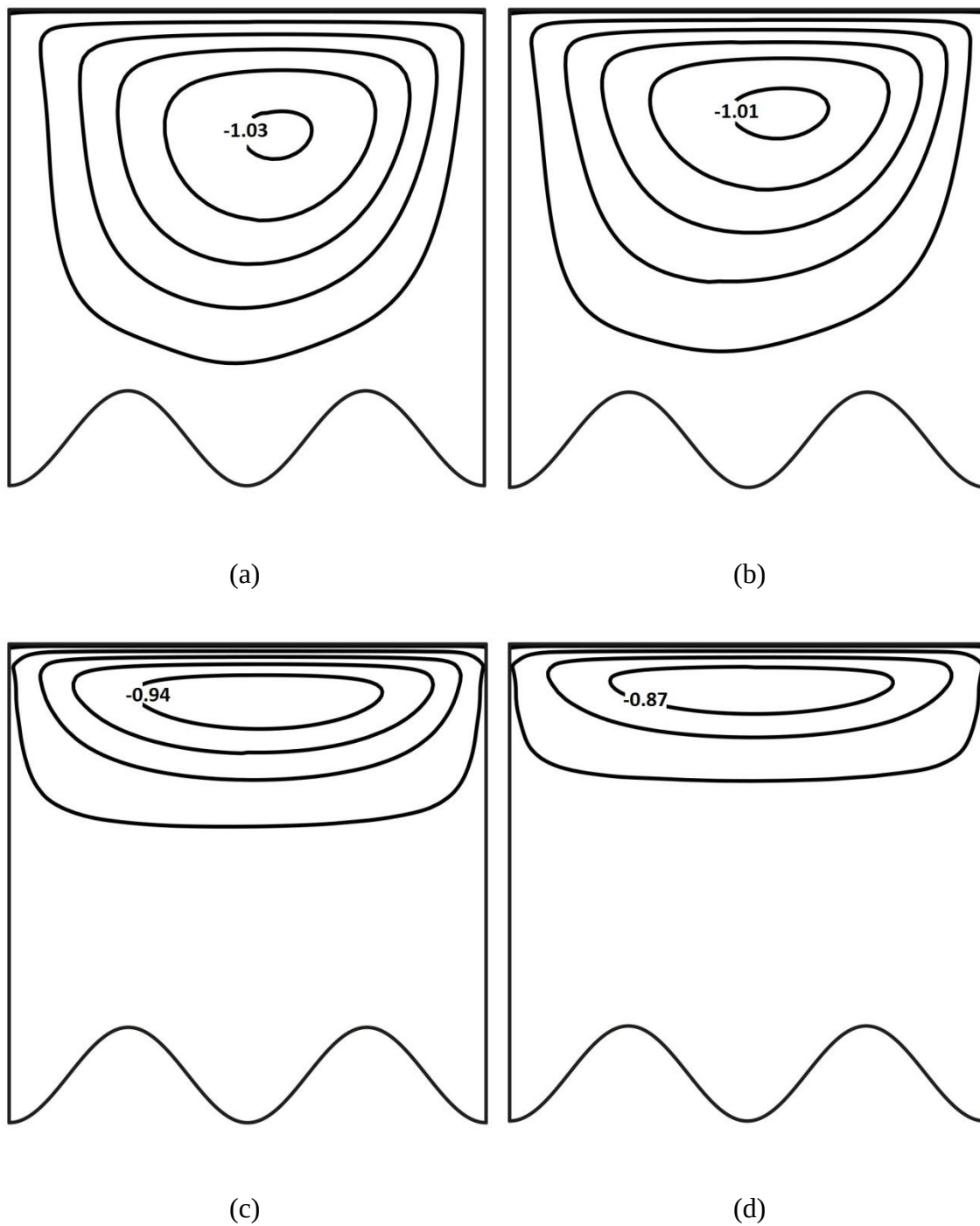


FIGURE 5.5: Variation of Streamlines for $\lambda = 0.10$ and $\varphi = 0.09$ at Different Hartmann Numbers;

(a) $Ha = 0$, (b) $Ha = 10$, (c) $Ha = 50$, (d) $Ha = 90$

5.5 Effect of Different Hartmann Numbers on Isotherms

Study of Hartmann number effect on isotherms shows a more clear indication of magnetic field effect on cavity heat transfer. Effects of increasing Hartmann numbers on isotherms was studied for a constant $\phi = 0.09$ and $Ra = 10^6$. Magnetic field tends to increase conductive effect over convective effect. For $Ha = 0$, no magnetic field is applied, so the isotherms on [Figure 5.6](#) shows normal isotherm causes by temperature difference and driven lid. Isotherms are clustered near the heated bottom wall. But with the increment of Ha , as the magnetic field starts to increase conduction starts to become more dominant. [Figure 5.6](#) shows that for higher Ha , clustered isotherms at the bottom wall bring out more spacing between them. Isotherms become almost parallel to each other. Conduction line distribution pattern is formed. In fact, at high Hartmann number flow reached quasiconduction regime. Convection is almost surpassed there.

5.6 Effect of Different Hartmann Numbers on Average Nusselt Number

The isotherms investigation is backed up by the study of average Nusselt number with Hartmann number. [Figure 5.7](#) indicates the effect of Hartmann number on average Nusselt number for various λ . And as predicted, results show that with the increment of Ha , Nu decreases. As Nusselt number stands for the ratio of convection versus conduction heat transfer, decreasing Nusselt number indicates domination of conduction. And at very high Hartmann number inclusion Nu becomes so low that conclusion can be drawn as, heat transfer is only done by conduction.

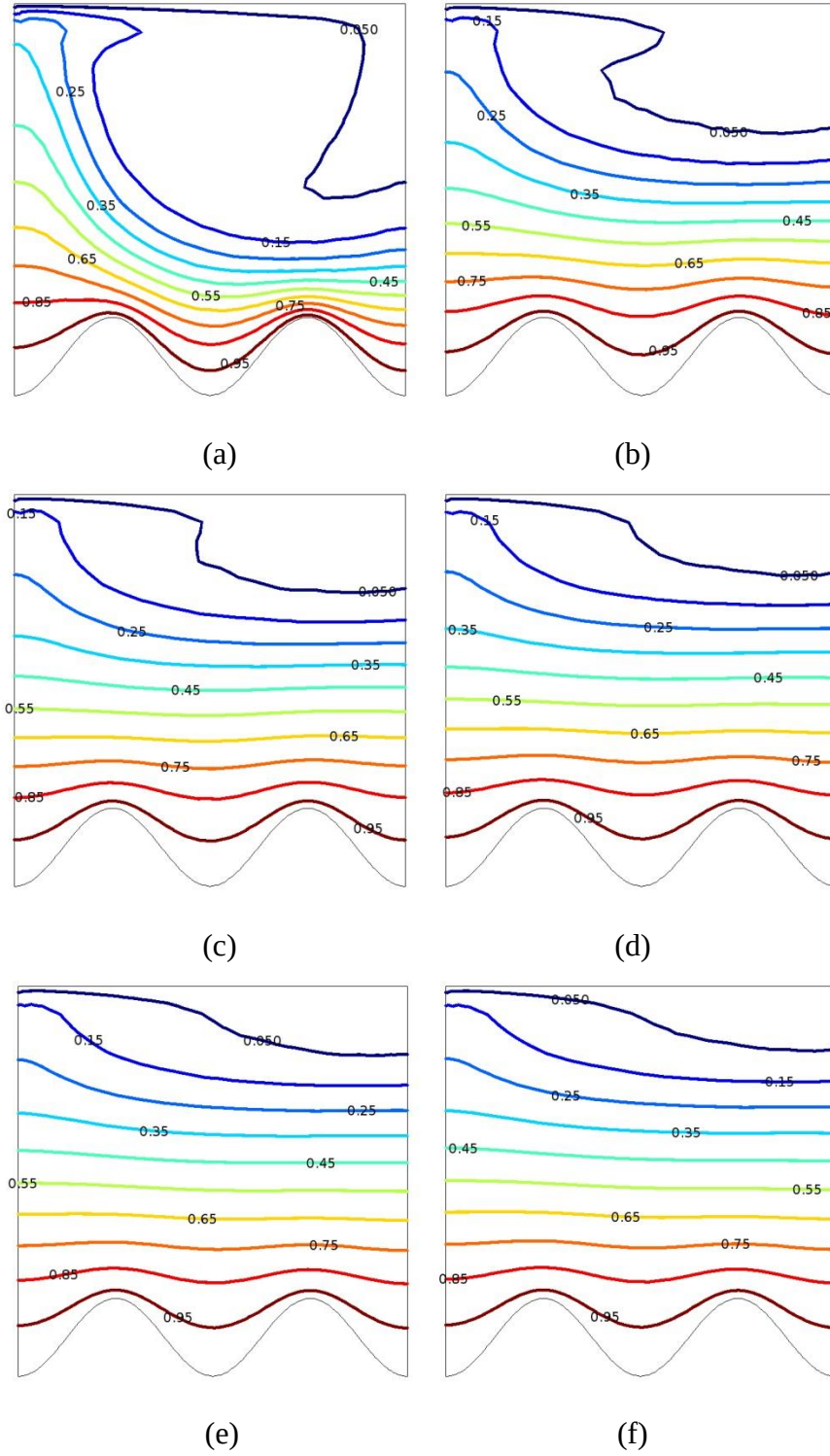


FIGURE 5.6: Variation of Isotherms for $\lambda = 0.10$ and $\phi = 0.09$ at Hartmann Numbers –
 (a) $Ha = 10$, (b) $Ha = 30$, (c) $Ha = 50$, (d) $Ha = 70$, (e) $Ha = 90$ and (f) $Ha = 100$

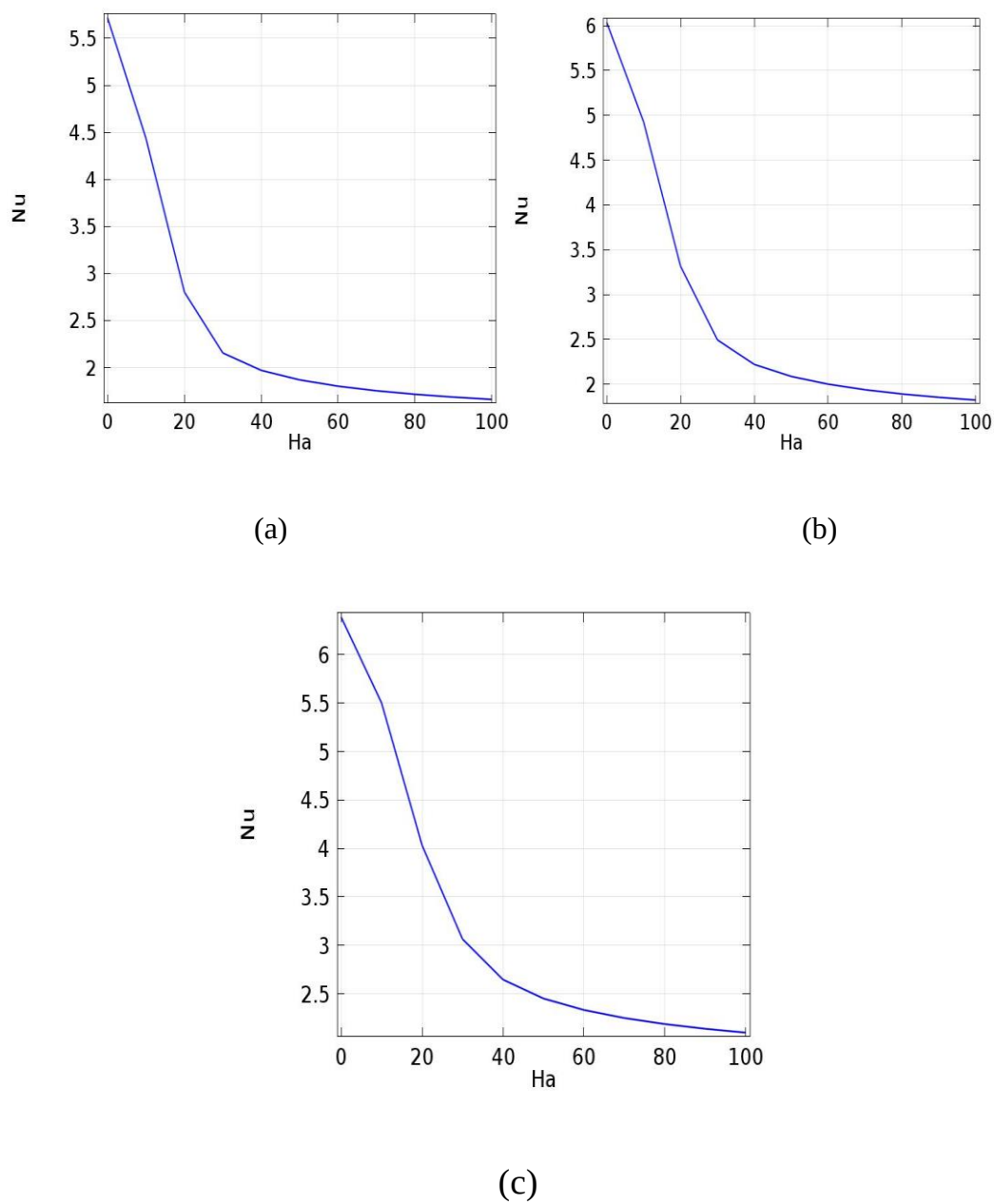


FIGURE 5.7: Variation of Average Nusselt Number for Different λ with Different Hartmann Numbers (a) $\lambda = 0.05$, (b) $\lambda = 0.10$, (c) $\lambda = 0.15$

5.7 Effect of Different Darcy Numbers on Streamlines

Darcy number (Da) corresponds to permeability of a media hence porosity. Increasing Darcy number means increasing permeability, so high porosity. Effect of Da was studied on streamlines for $Ha = 50$, $Ra = 10^6$, $Ri = 1$ and $\varphi = 0.09$.

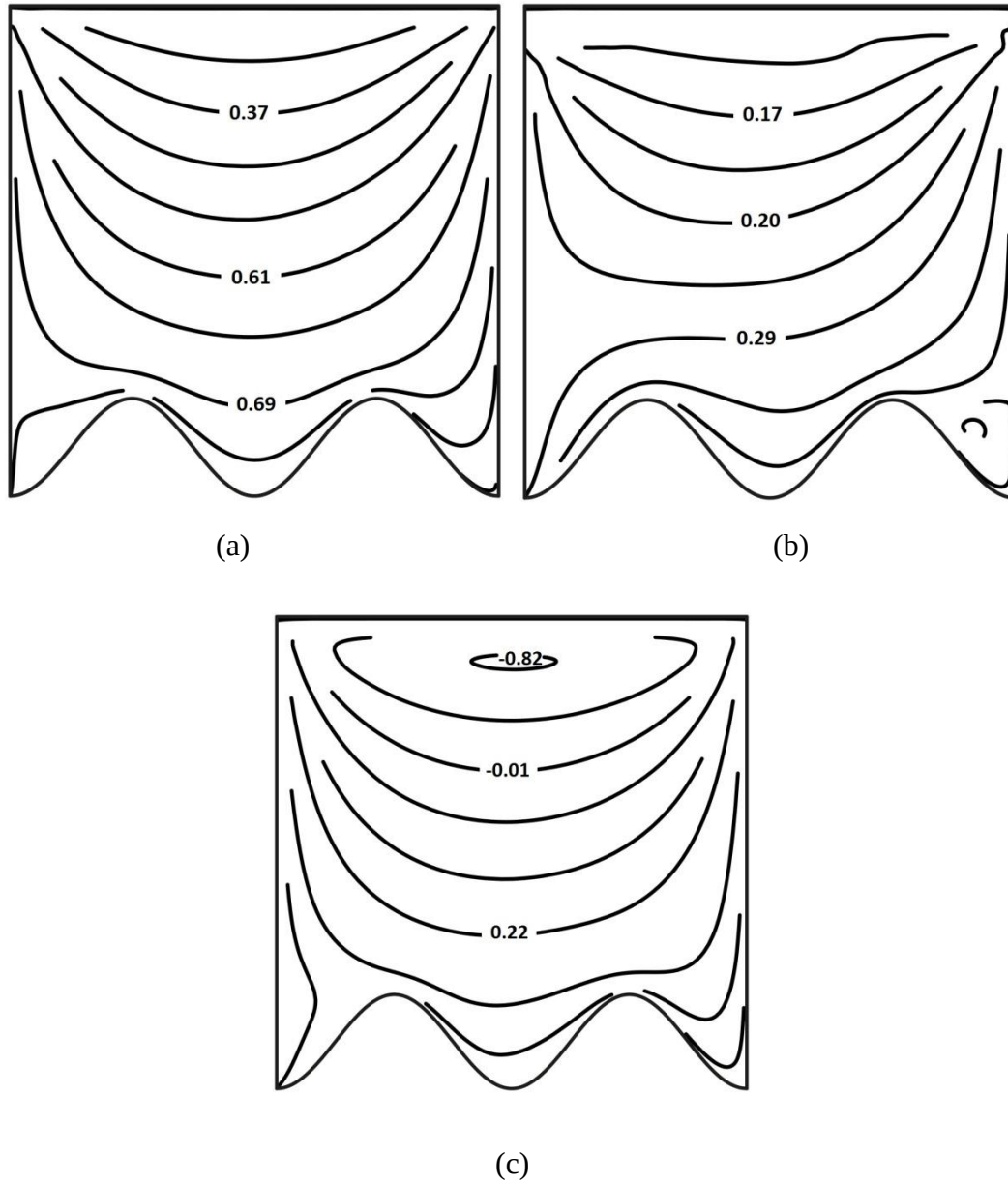


FIGURE 5.8: Variation of Streamlines for $\lambda = 0.10$ at Different Darcy Numbers where $Ha = 50$, $Ra = 10^6$, $Ri = 1$ and $\varphi = 0.09$;

(a) $Da = 0.00001$, (b) $Da = 0.0001$, (c) $Da = 0.001$.

With the increment of Da , circulation becomes stronger. So while at lower Da , flow has no eddy or circular pattern, higher Da tends to form an eddy in the cavity. It can be seen from [figure 5.8](#) where at $Da = 0.00001$ flow has a very low strength that there is no core or eddy in the streamlines. But at $Da = 0.001$ flow gains sufficient strength to form an eddy flow with a core. Studies were done with the assumption that CNT dimensions are small enough to circulate through a porous media. So, single phase study was done.

5.8 Summary

Amplitude of the curvature of heated wall, Richardson number, solid volume fraction of CNT-water, Hartmann number, Darcy number have significant effect on heat transfer and flow velocity. Increasing Richardson number causes increment of Nusselt number. Also high solid volume fraction results in higher Nusselt number. Magnetic field reduces heat transfer and flow strength. As a result, increasing Hartmann number causes decrease in Nusselt number. And increment of Darcy number increases flow strength causing circular flow or eddy.

CHAPTER 6

Conclusion and Future Prospects

Problems of steady, laminar mixed convection inside a lid driven cavity with curved bottom walls are studied by varying Richardson number, Hartmann number and Darcy number along with the variation of the amplitude of curved bottom wall. CNT-water nanofluid was used with a solid volume fraction of Carbon nanotube up to 9%.

6.1 Concluding Remarks

Following concluding remarks can be drawn from the computational analysis of the present problem:

- Inclusion of nanoparticles (CNT) in base fluid signifies augmentation in heat transfer performance.
- As Richardson number increases, heat transfer rate increases significantly.
- Velocity profile changes drastically for higher Richardson number and maximum amplitude of the wavy bottom wall.
- Better heat transfer is observed for maximum amplitude of wavy bottom wall.
- Applying magnetic field reduces Nusselt number, therefore reduces heat transfer rate.
- Strength of the flow field decreases with increasing Hartman number.
- Strength of circulation increases with increasing Darcy number.

6.2 Future Prospects

In this study heat transfer phenomena in a lid driven cavity was studied for CNT-water nanofluid. Magnetic field and porous medium are also included later. A very meticulous view of flow and thermal characteristics is presented in the thesis. The boundary conditions and diversity are considered here are quite new and unique. So still a lot of opportunity waits for this case to be studied, both numerically and experimentally. Some of the future prospects can be:

- Inclusion of mass transfer along with heat transfer.
- Observation of entropy generation.
- Further investigation changing the boundary condition & flow characteristics.
- Comparison between different nanofluids can be studied for the same cases.
- Study can be done using multilayer CNT or changing the dimensions of the nanotubes.
As different dimension provide different heat transfer phenomena.
- Present work was done by for single phase assumption. Double phase studies can be done. This can specially provide effective results for porous media.
- Cavity shape can be changed and also inclusion of fins or different geometries inside the cavity can be considered.
- Present work has been carried out for steady state. So a further investigation can be done using transient flow.
- The study can be done experimentally so that a more realistic investigation can be done which can effectively be put into real applications.

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